Chapter 29

Cryptography

29.1 Introduction

Introduction to Cryptography

Figure 29.1 Cryptography components

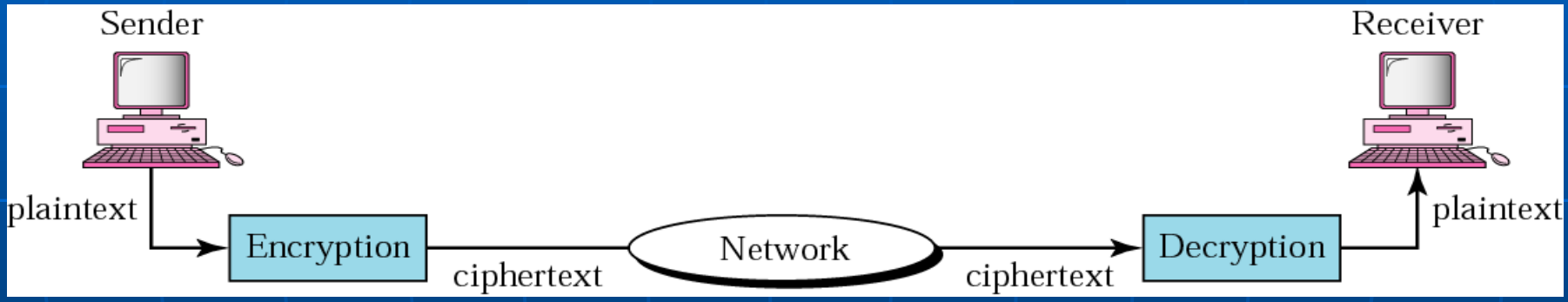
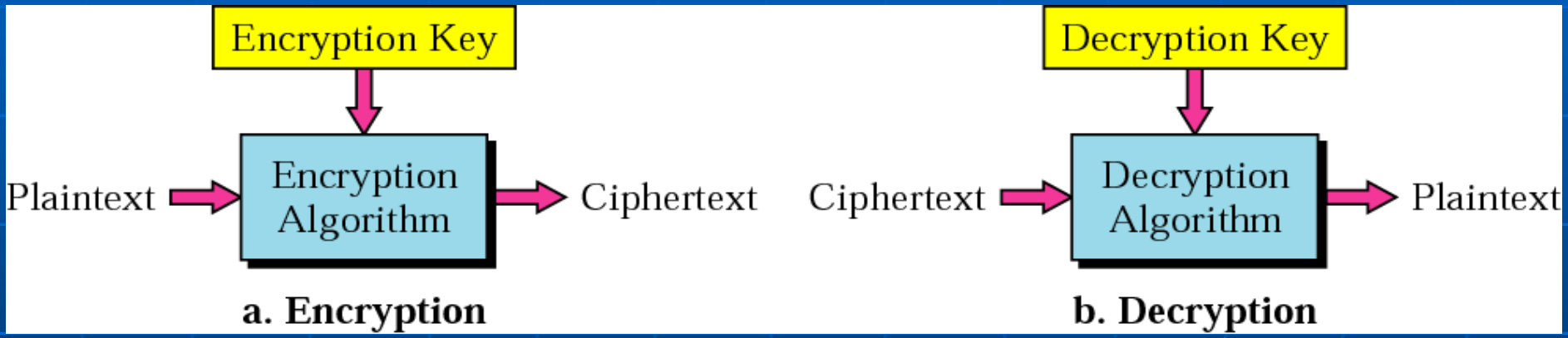


Figure 29.2 Encryption and decryption





Note:

In cryptography, the encryption/decryption algorithms are public; the keys are secret.

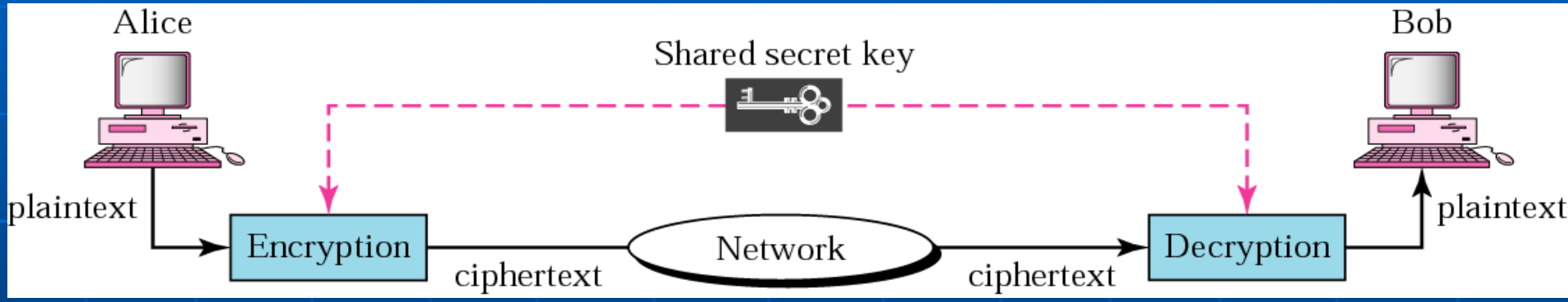
29.2 Symmetric-Key Cryptography

Traditional Cipher

Block Cipher

Operation Modes

Figure 29.3 Symmetric-key cryptography





Note:

In symmetric-key cryptography, the same key is used by the sender (for encryption) and the receiver (for decryption). The key is shared.



Note:

In symmetric-key cryptography, the same key is used in both directions.



Note:

Symmetric-key cryptography is often used for long messages.

Figure 29.4 Caesar cipher

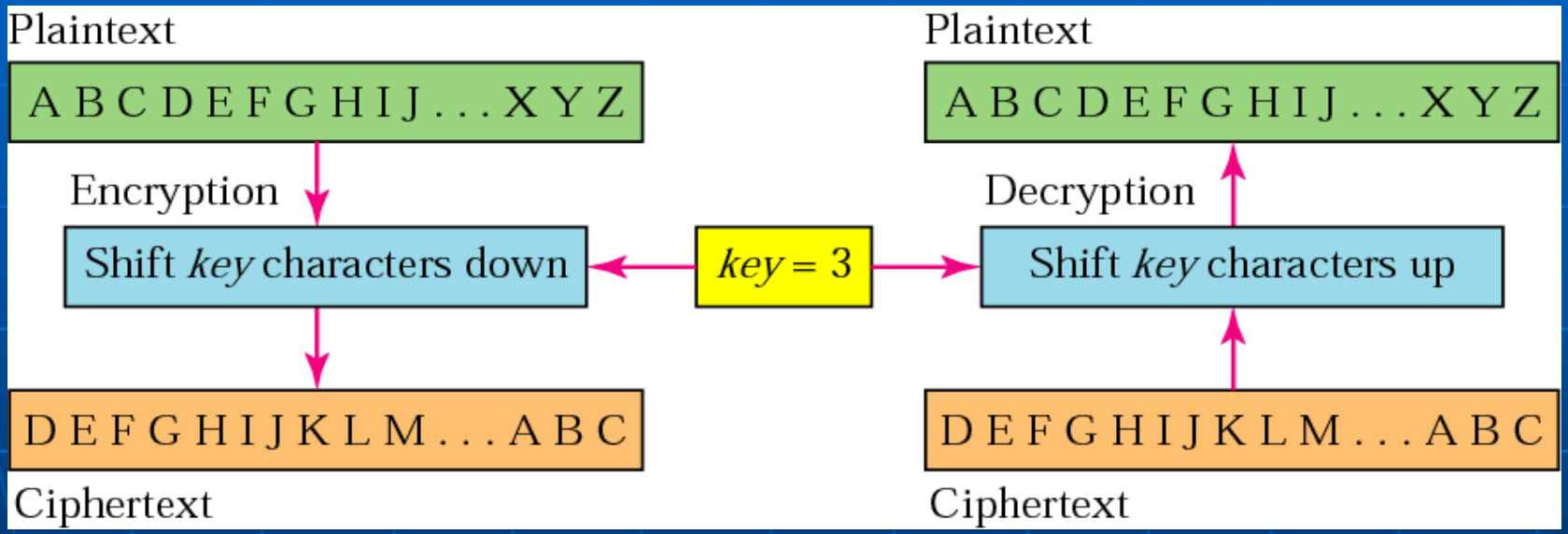
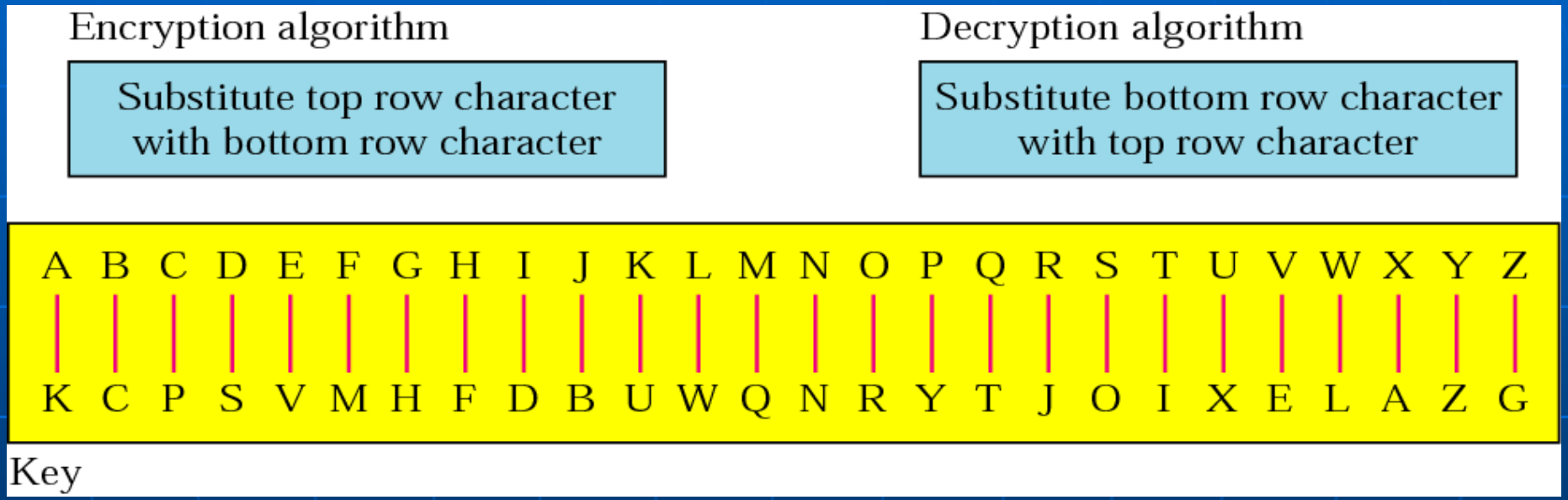


Figure 29.5 Example of monoalphabetic substitution

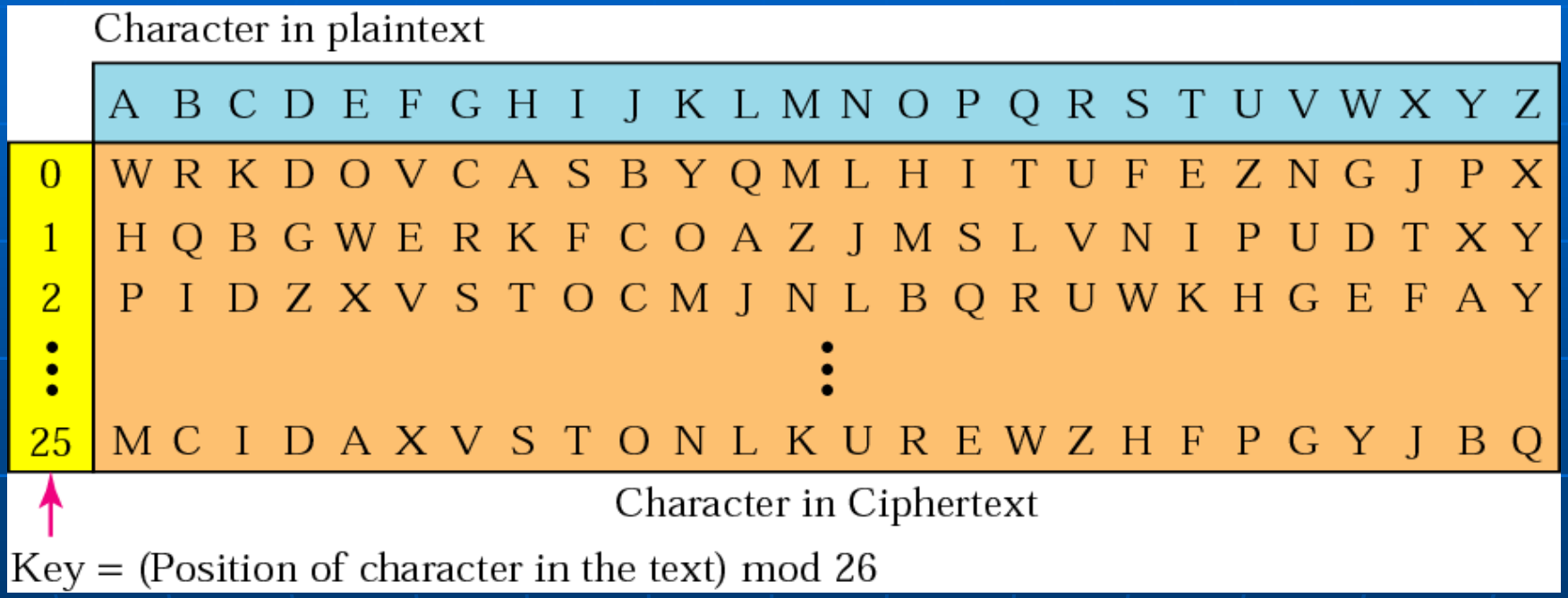




Note:

In monoalphabetic substitution, the relationship between a character in the plaintext to the character in the ciphertext is always one-to-one.

Figure 29.6 Vigenere cipher





Note:

In polyalphabetic substitution, the relationship between a character in the plaintext and a character in the ciphertext is one-to-many.

Figure 29.7 Transpositional cipher

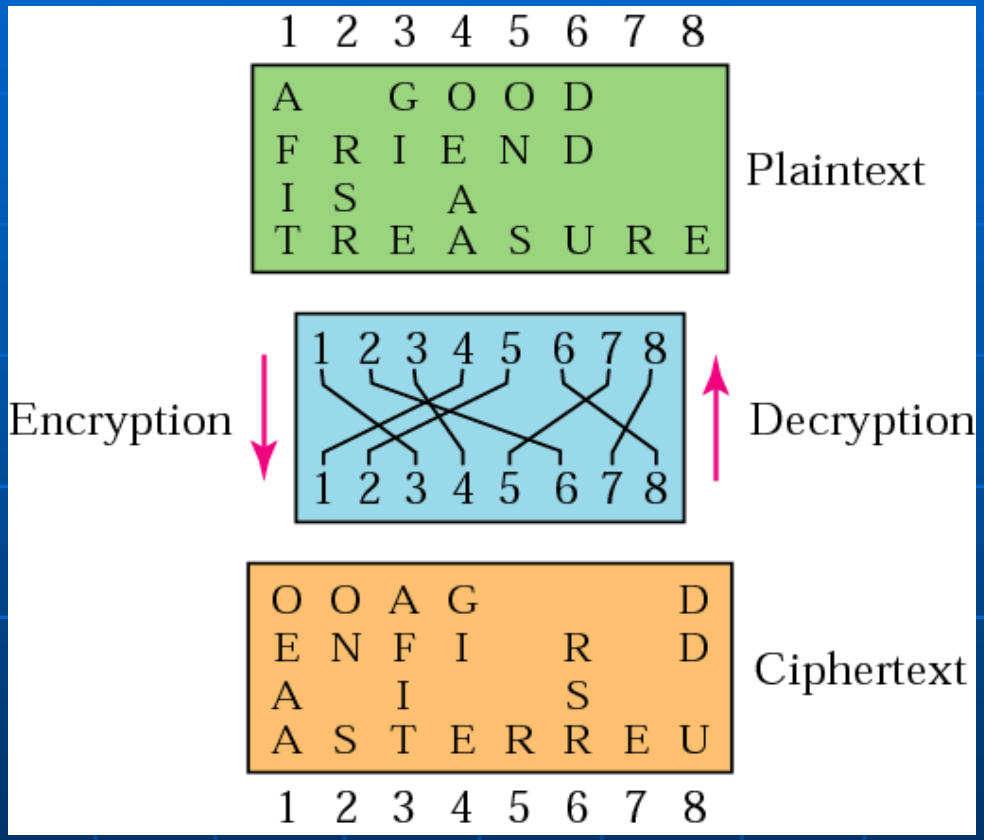


Figure 29.8 Block cipher

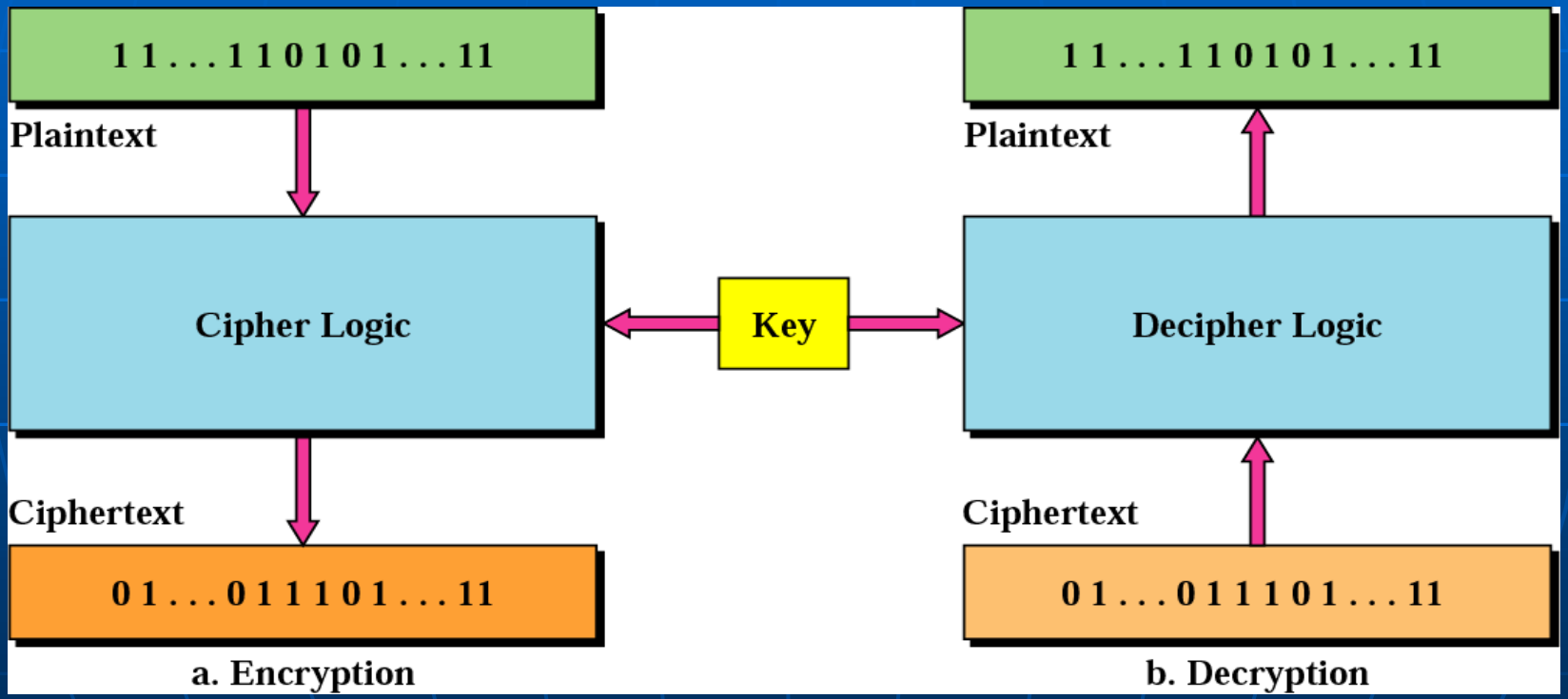


Figure 29.9 P-box

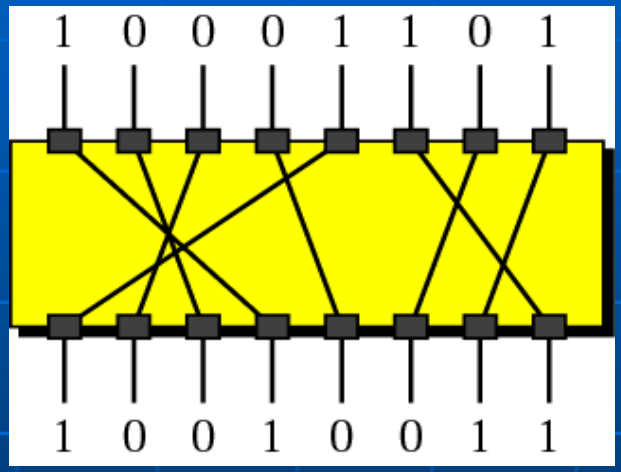


Figure 29.10 S-box

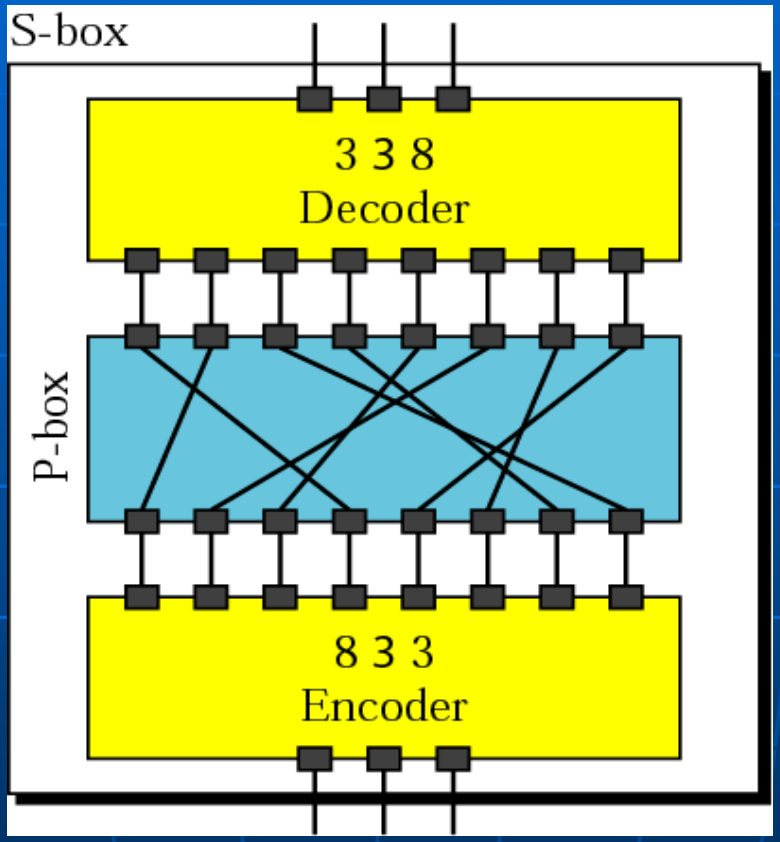


Figure 29.11 Product block

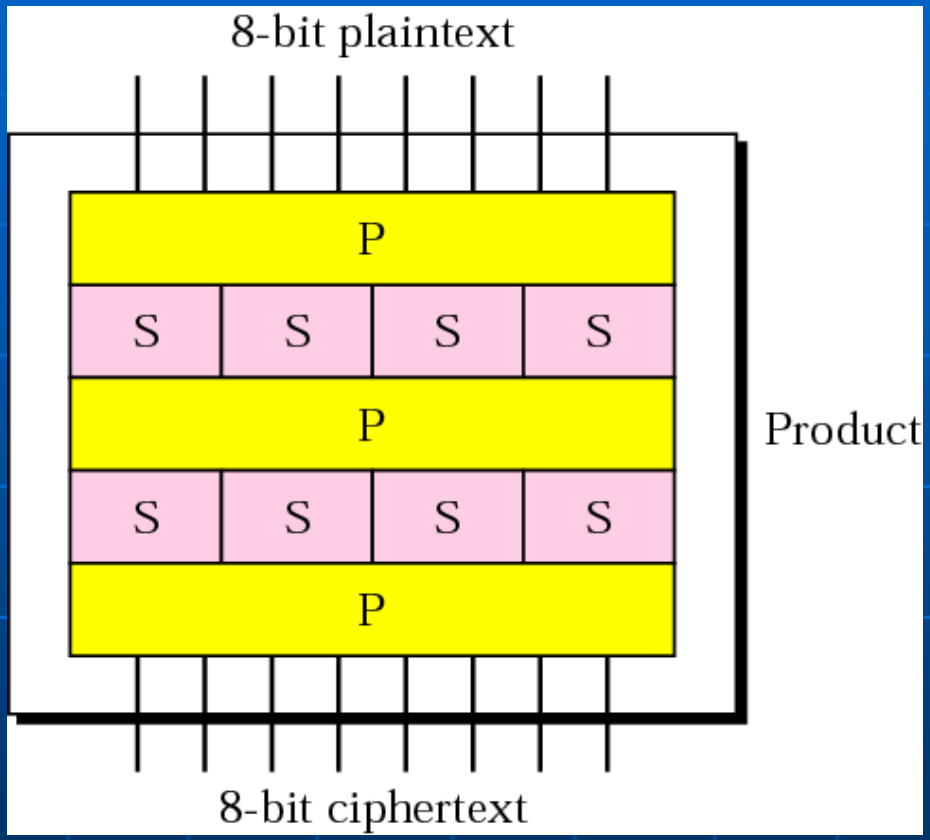


Figure 29.12 DES

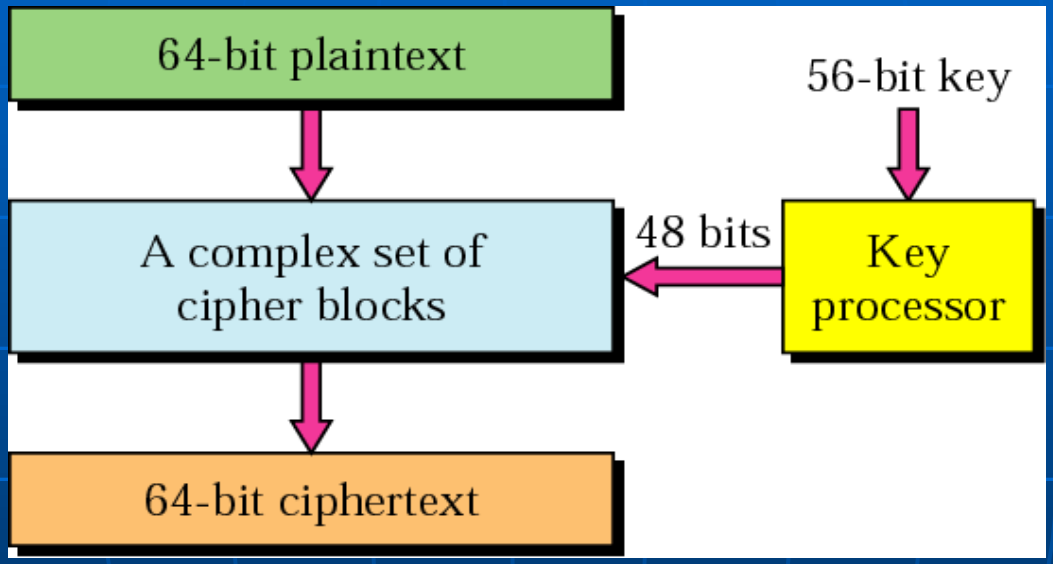


Figure 29.13 General scheme of DES

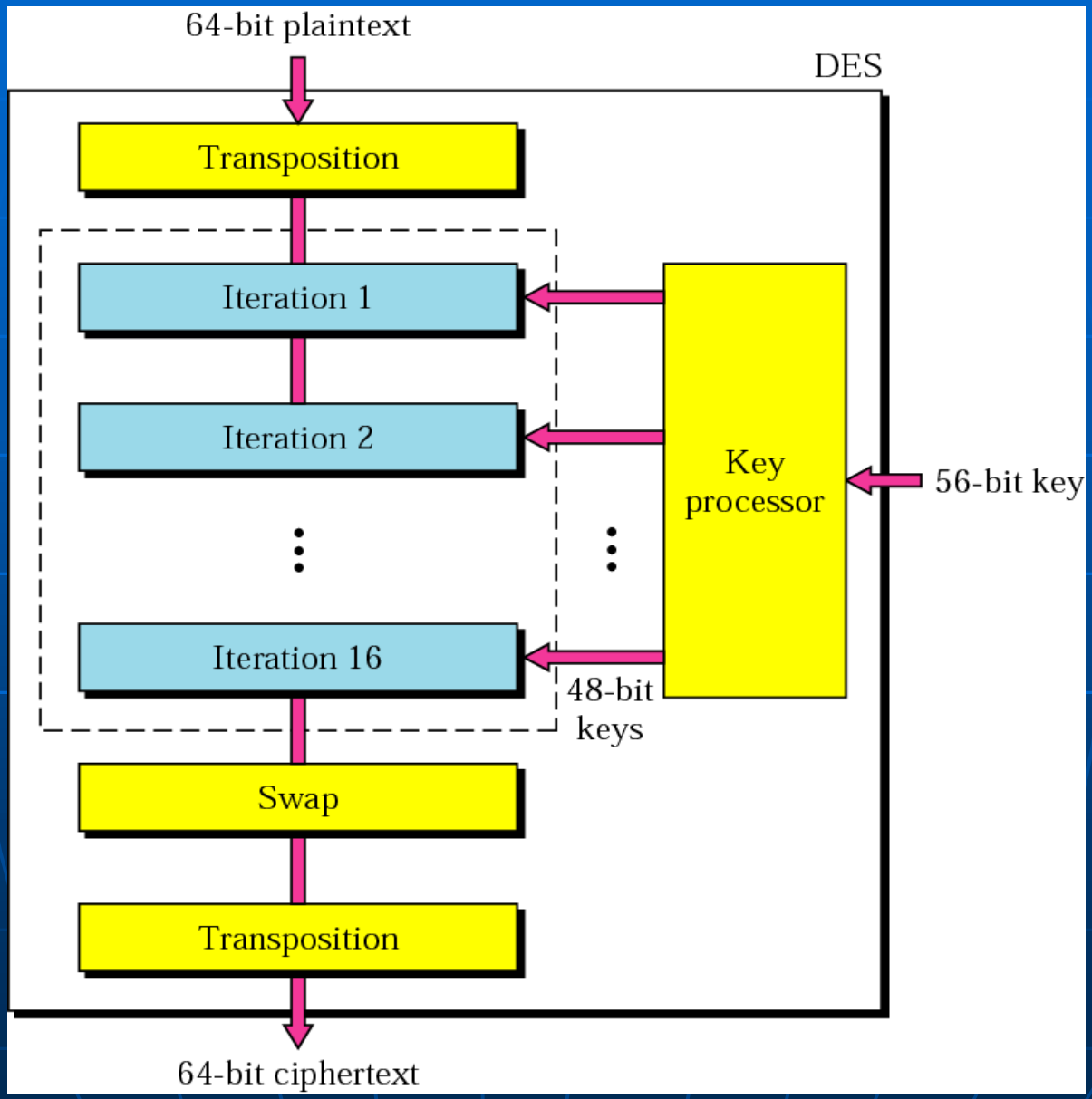


Figure 29.14 Iteration block

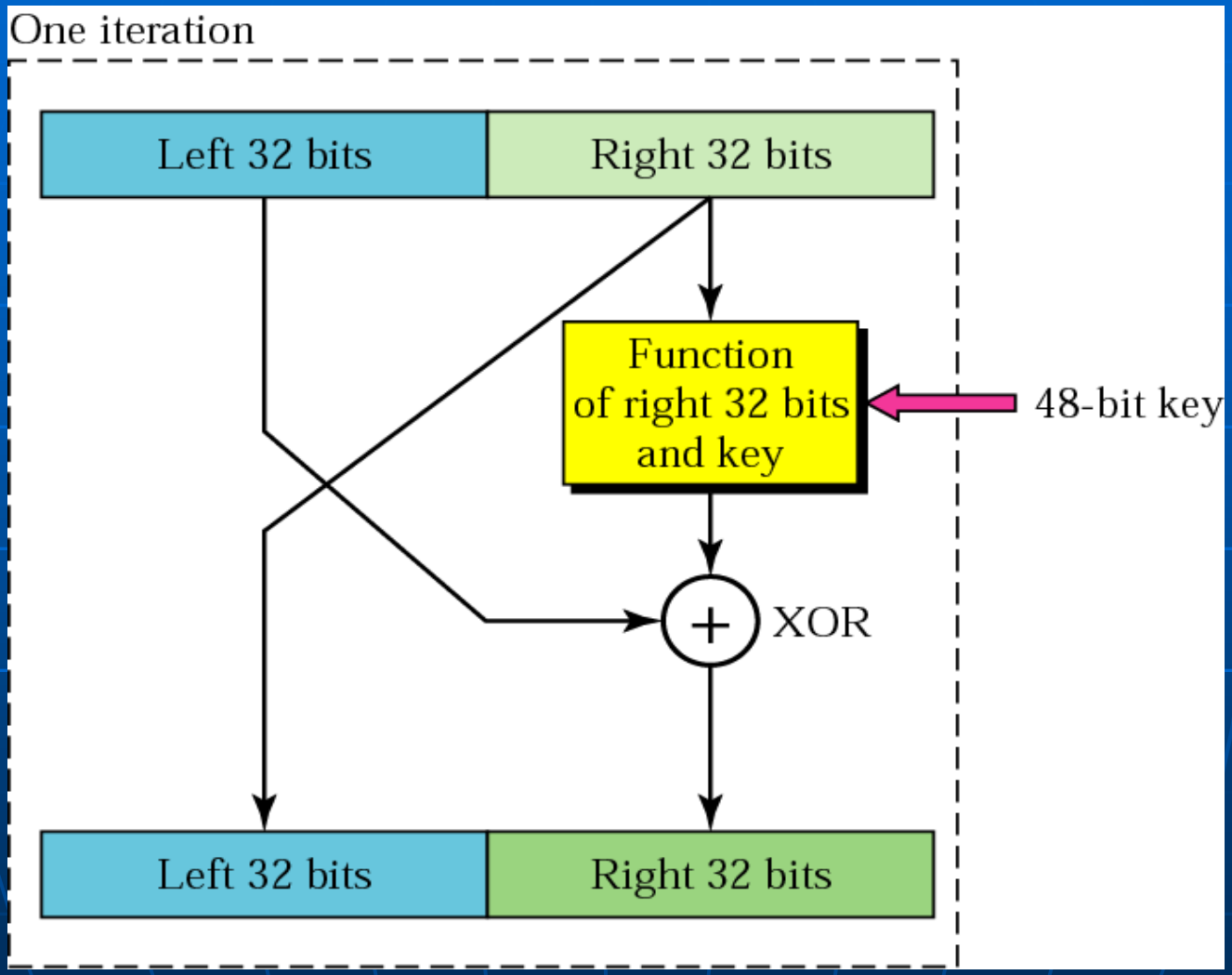
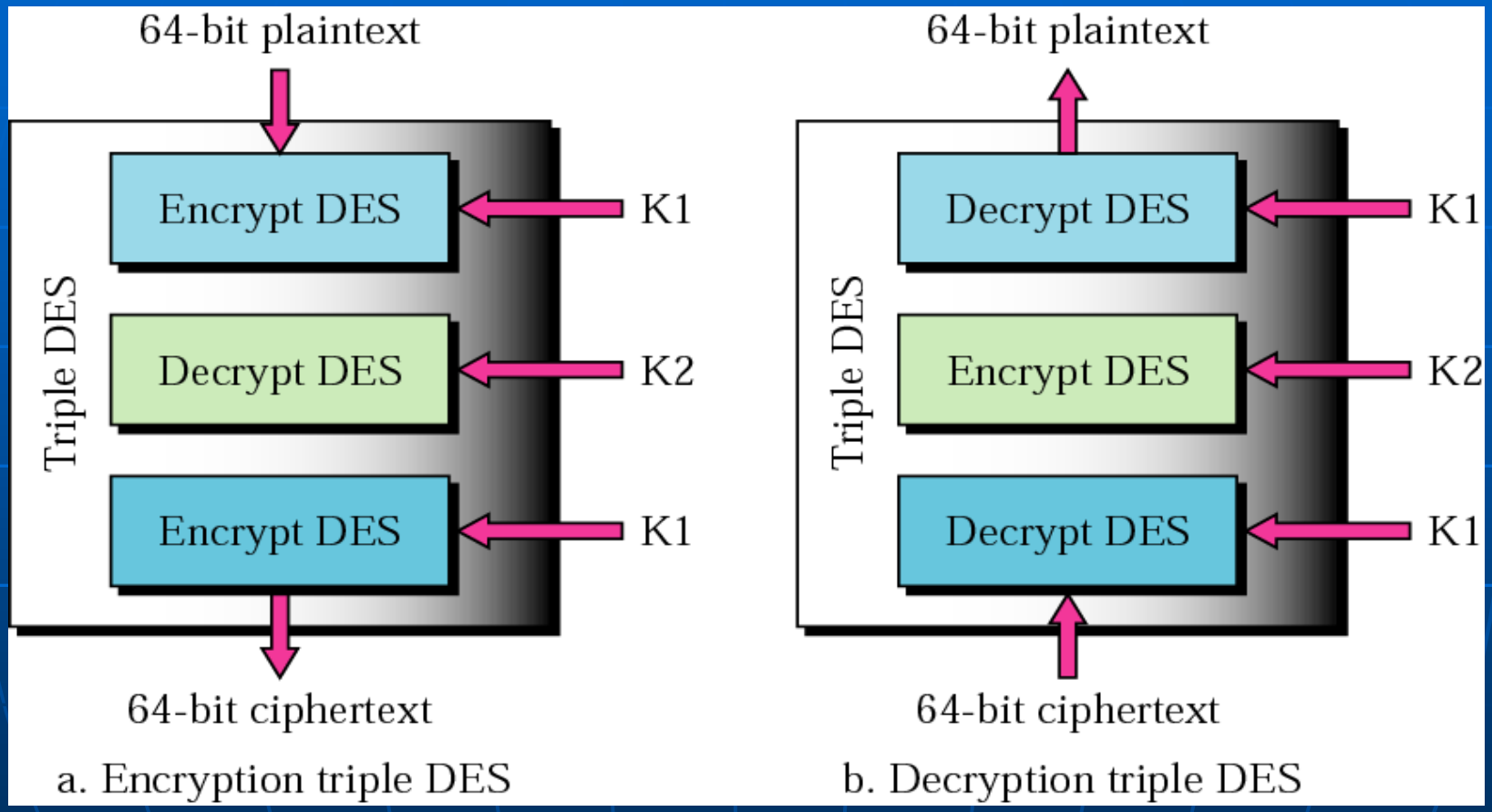


Figure 29.15 Triple DES





The DES cipher uses the same concept as the Caesar cipher, but the encryption/decryption algorithm is much more complex due to the sixteen 48-bit keys derived from a 56-bit key.

Figure 29.16 ECB mode

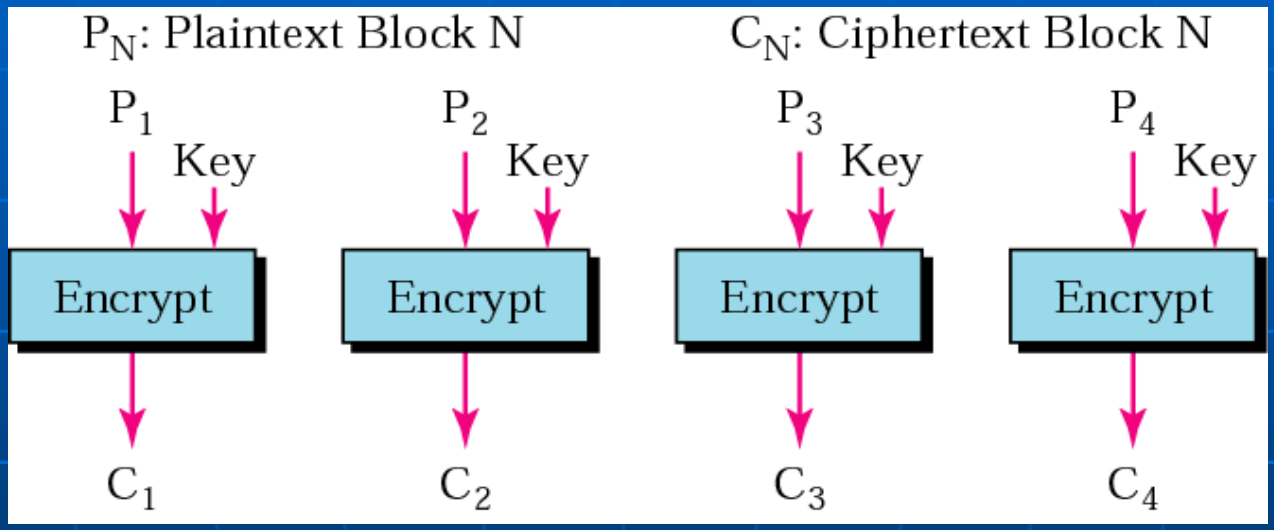


Figure 29.17 CBC mode

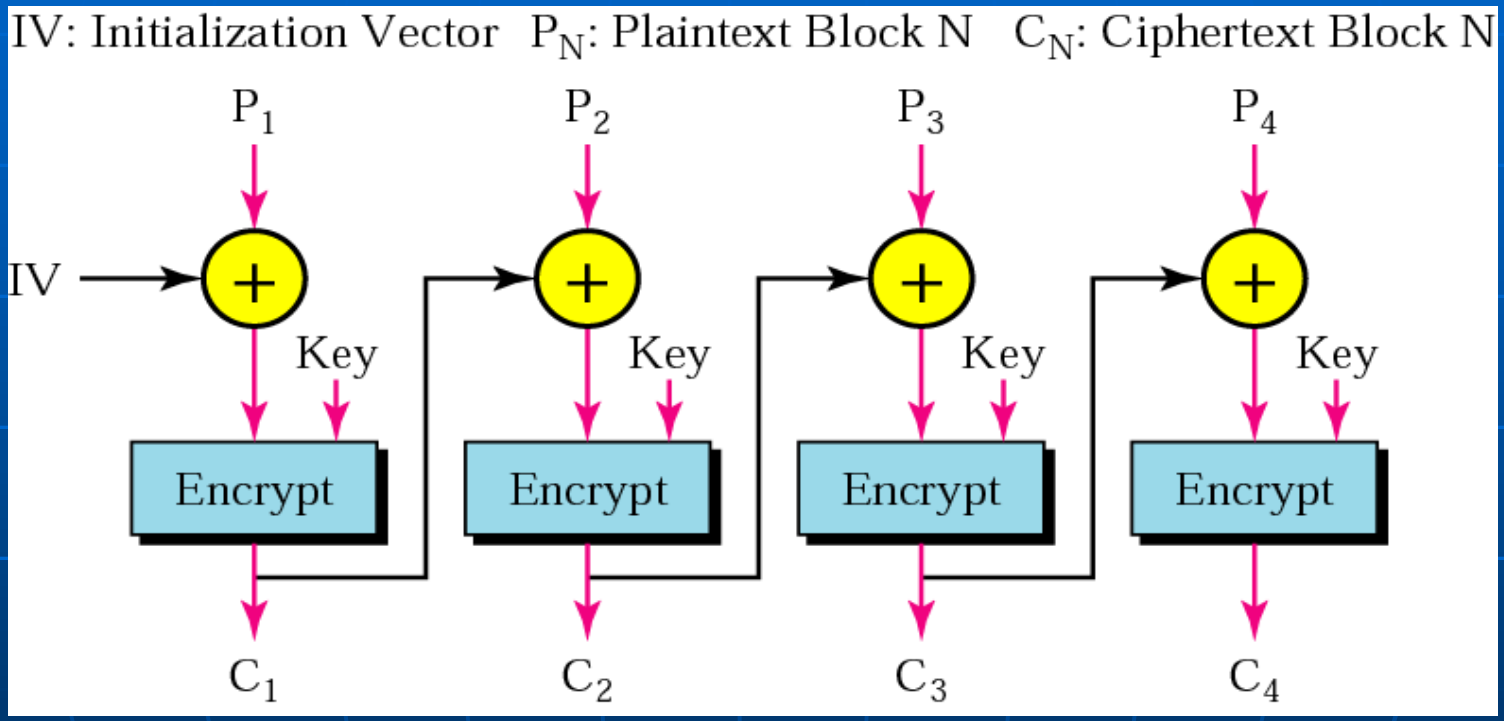


Figure 29.18 CFM

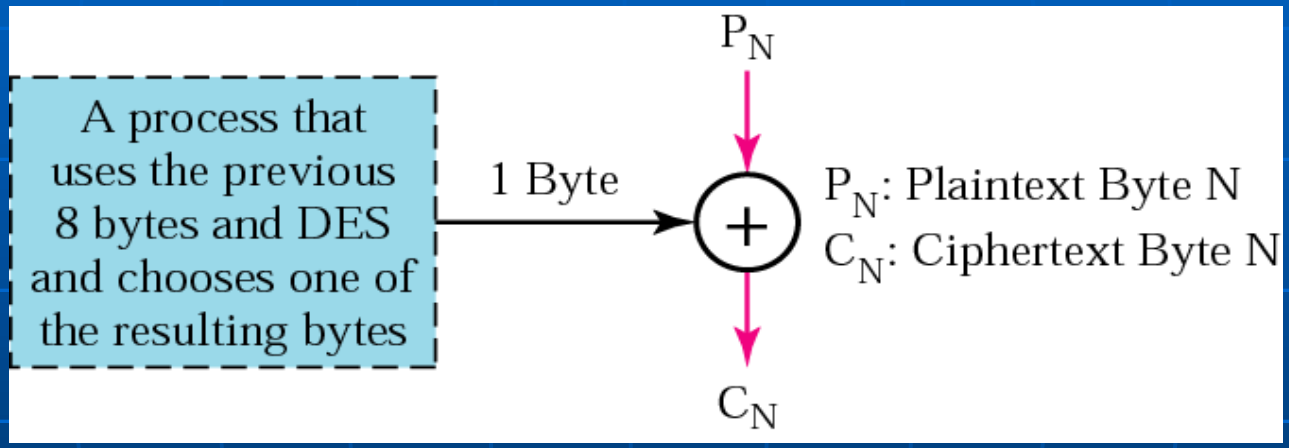
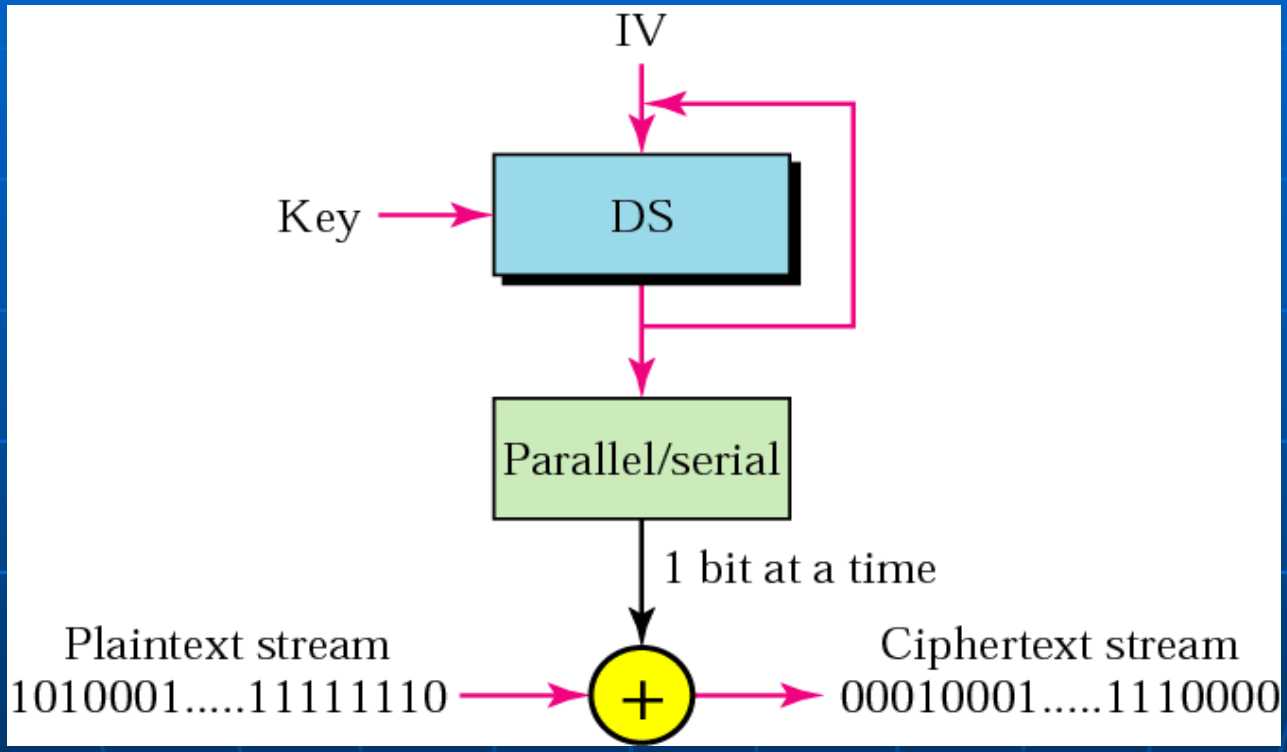


Figure 29.19 CSM

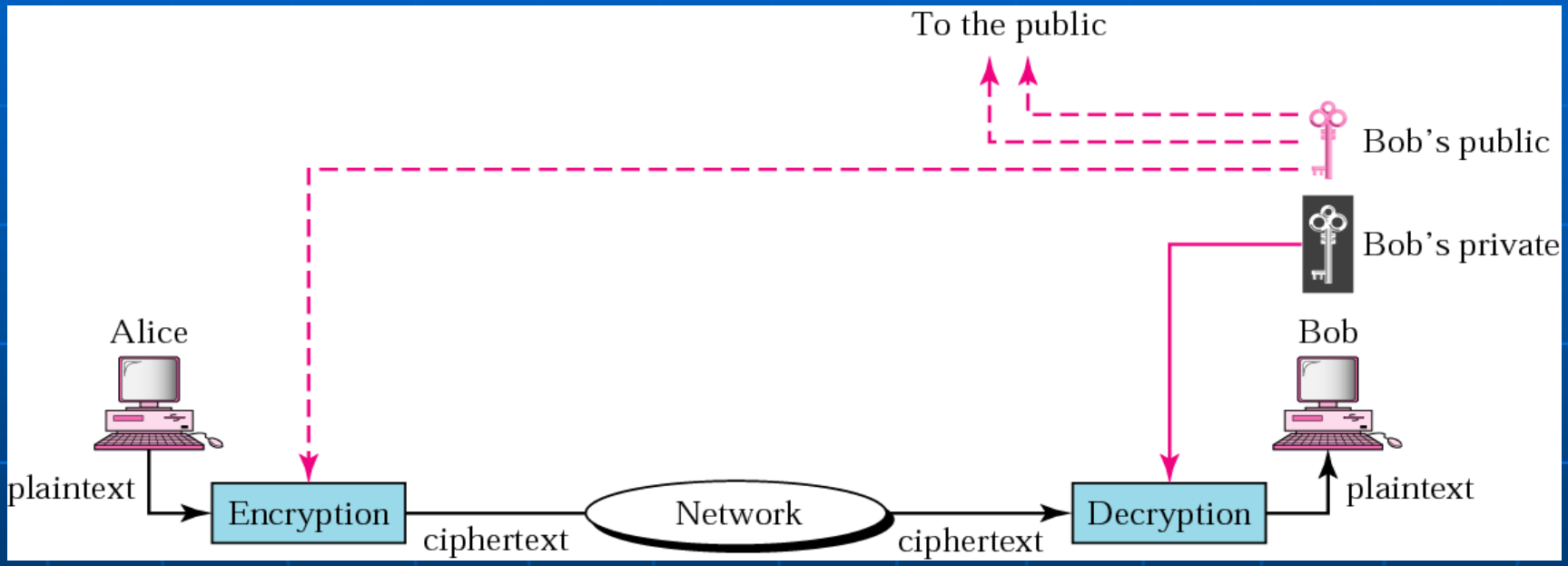


29.3 Public-Key Cryptography

RSA

Choosing Public and Private Keys

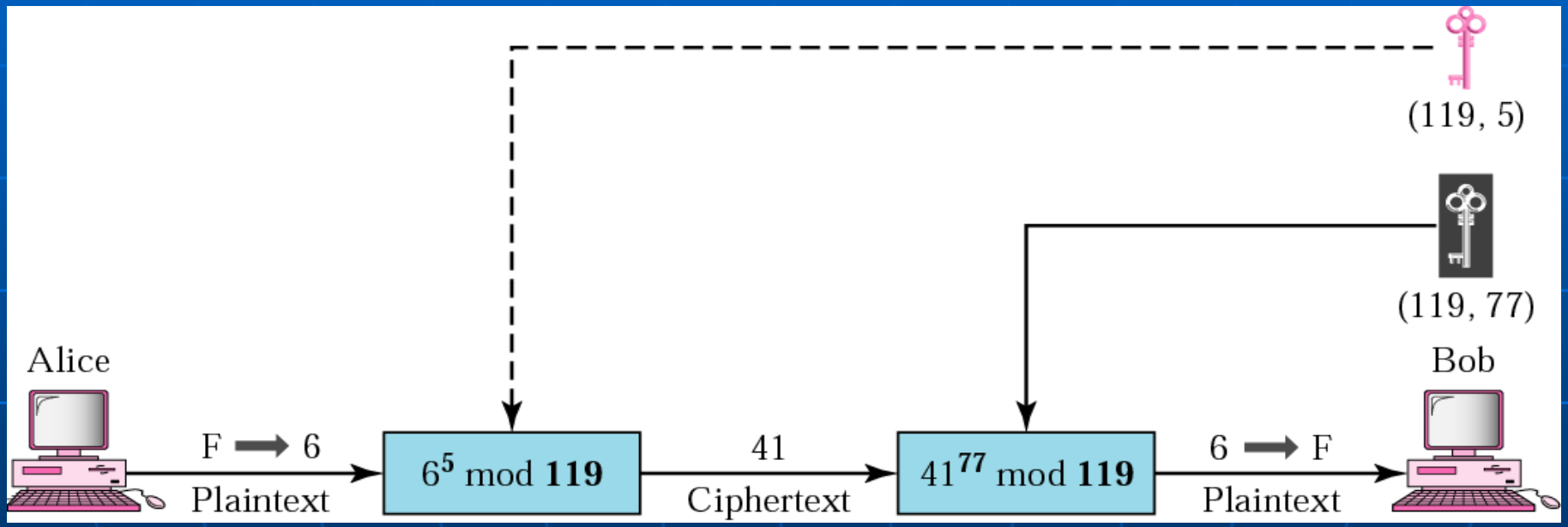
Figure 29.20 Public-key cryptography





Public-key algorithms are more efficient for short messages.

Figure 29.21 RSA



Chapter 30

Message Security, User Authentication, and Key Management

30.1 Message Security

Privacy

Authentication

Integrity

Nonrepudiation

Figure 30.1 Message security

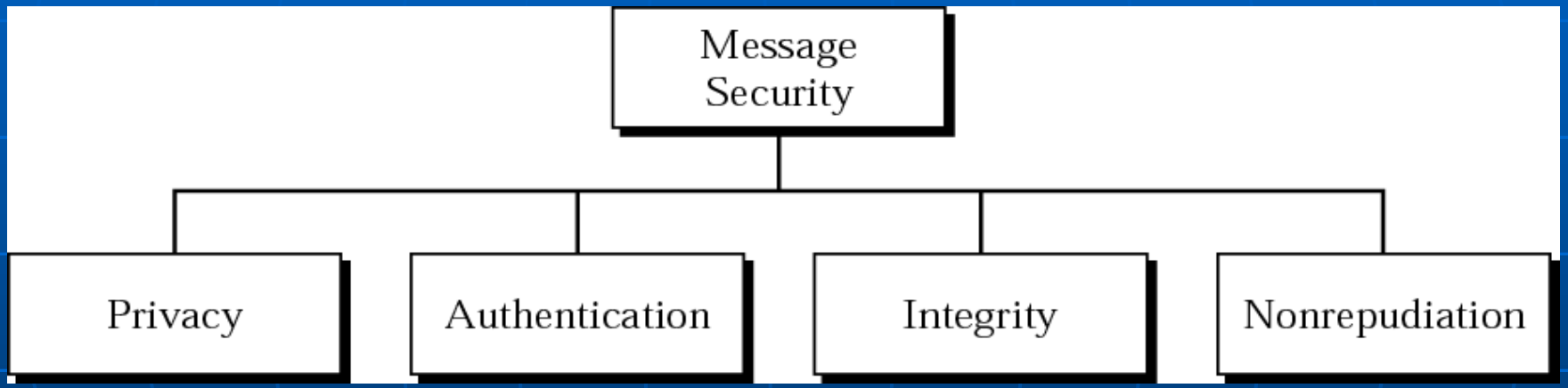


Figure 30.2 Privacy using symmetric-key encryption

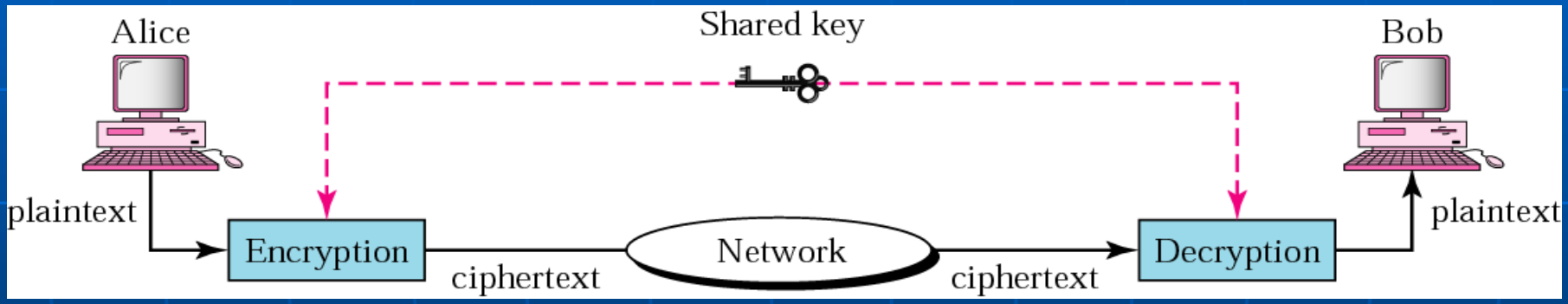
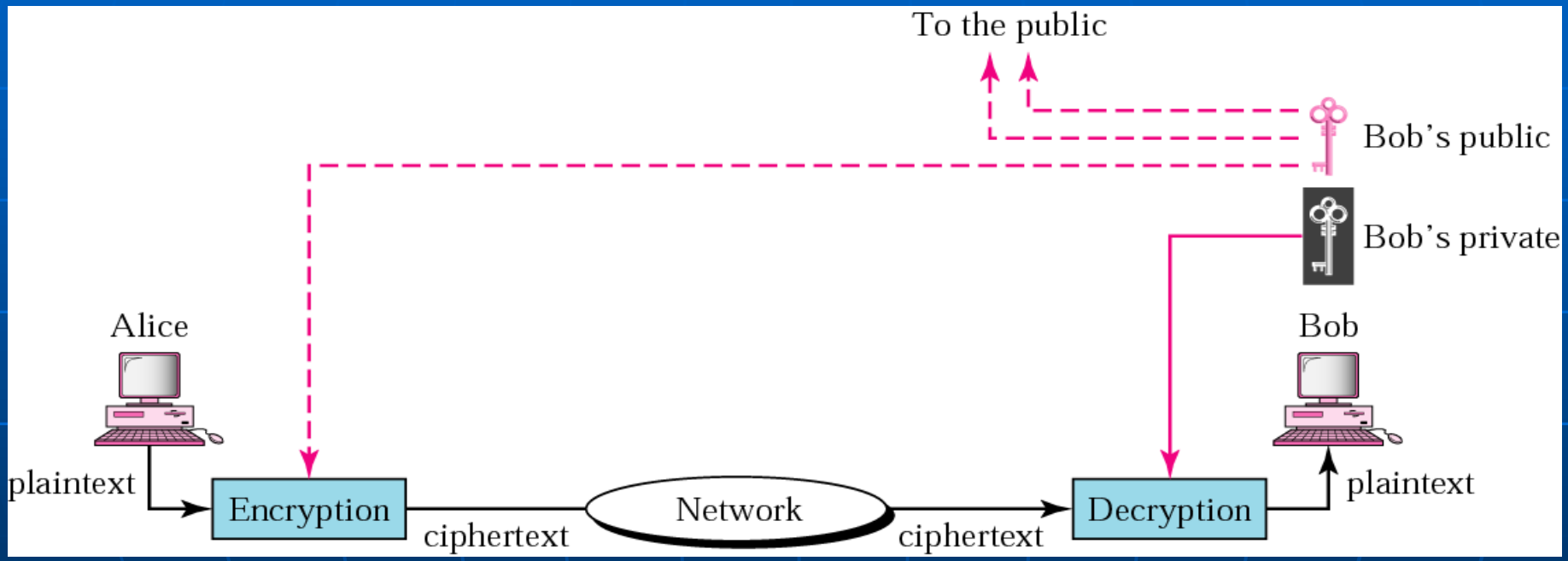


Figure 30.3 Privacy using public-key encryption

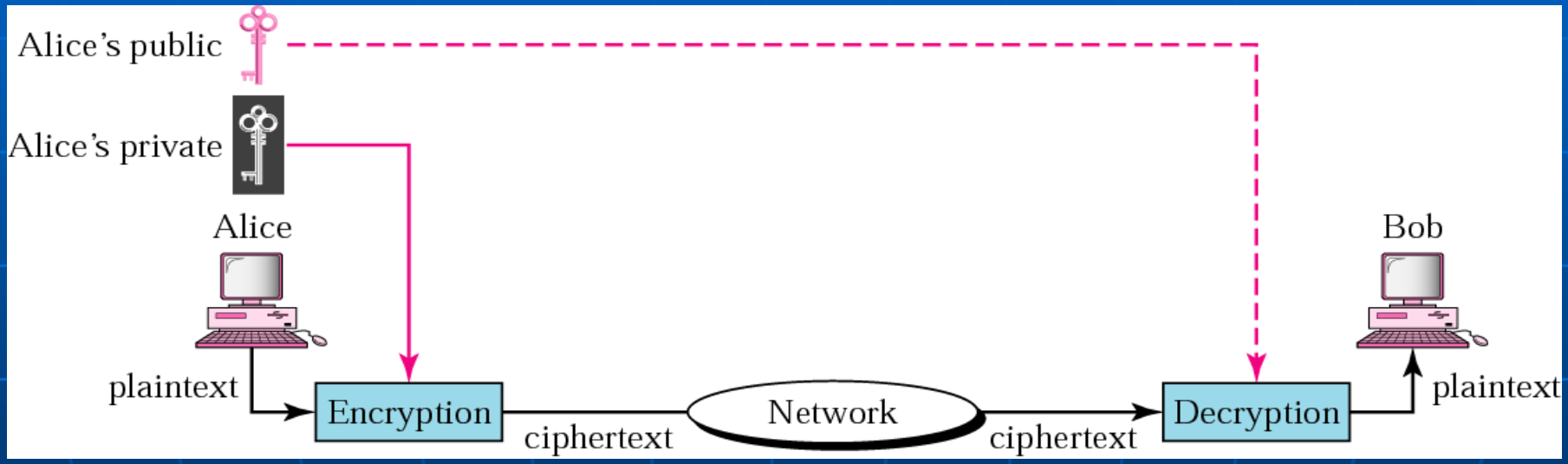


30.2 Digital Signature

Signing the Whole Document

Signing the Digest

Figure 30.4 Signing the whole document





Note:

Digital signature does not provide privacy. If there is a need for privacy, another layer of encryption/decryption must be applied.

Figure 30.5 Signing the digest

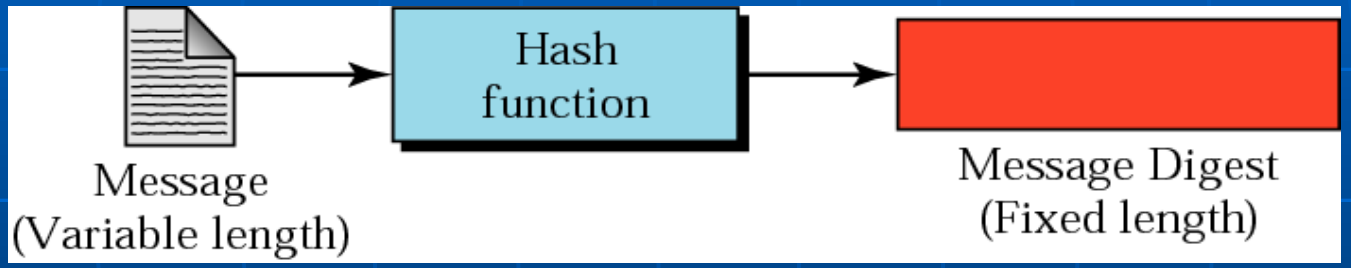


Figure 30.6 Sender site

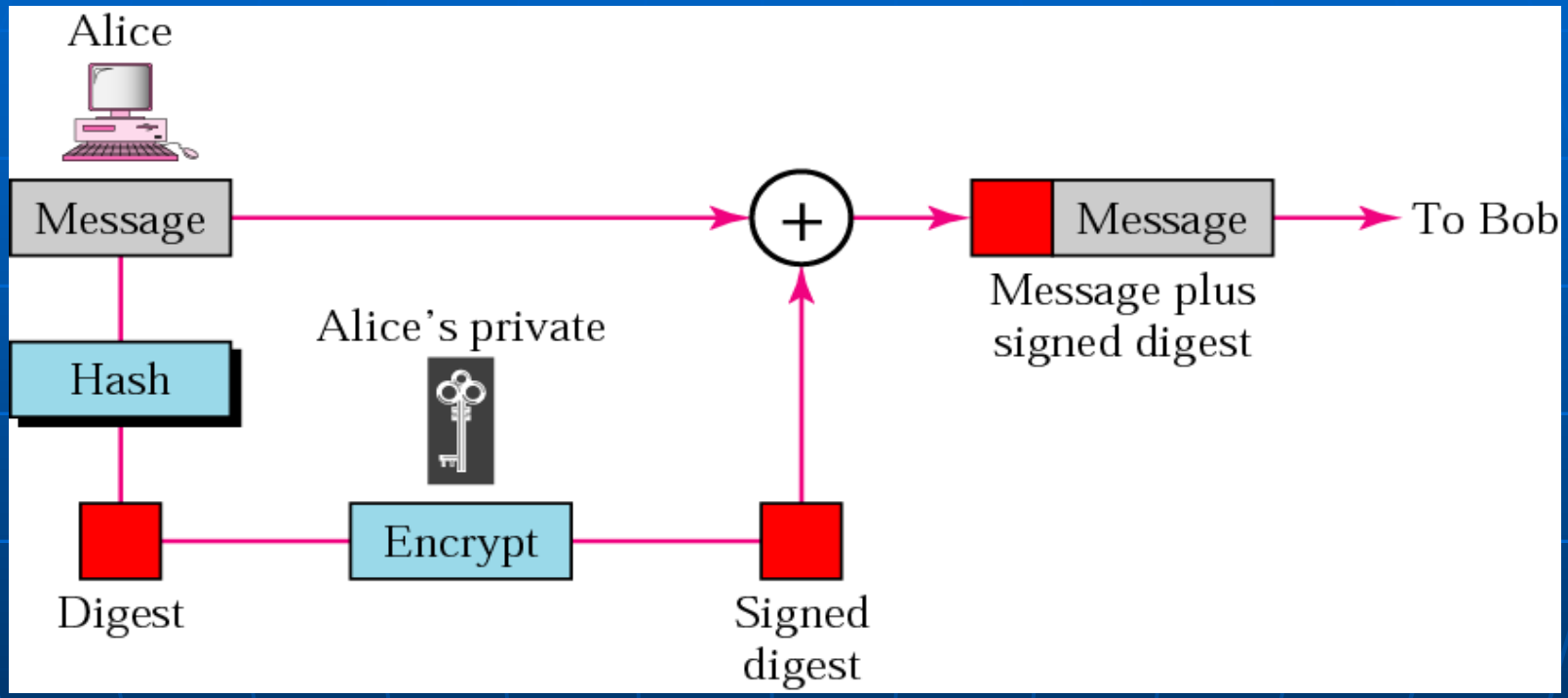
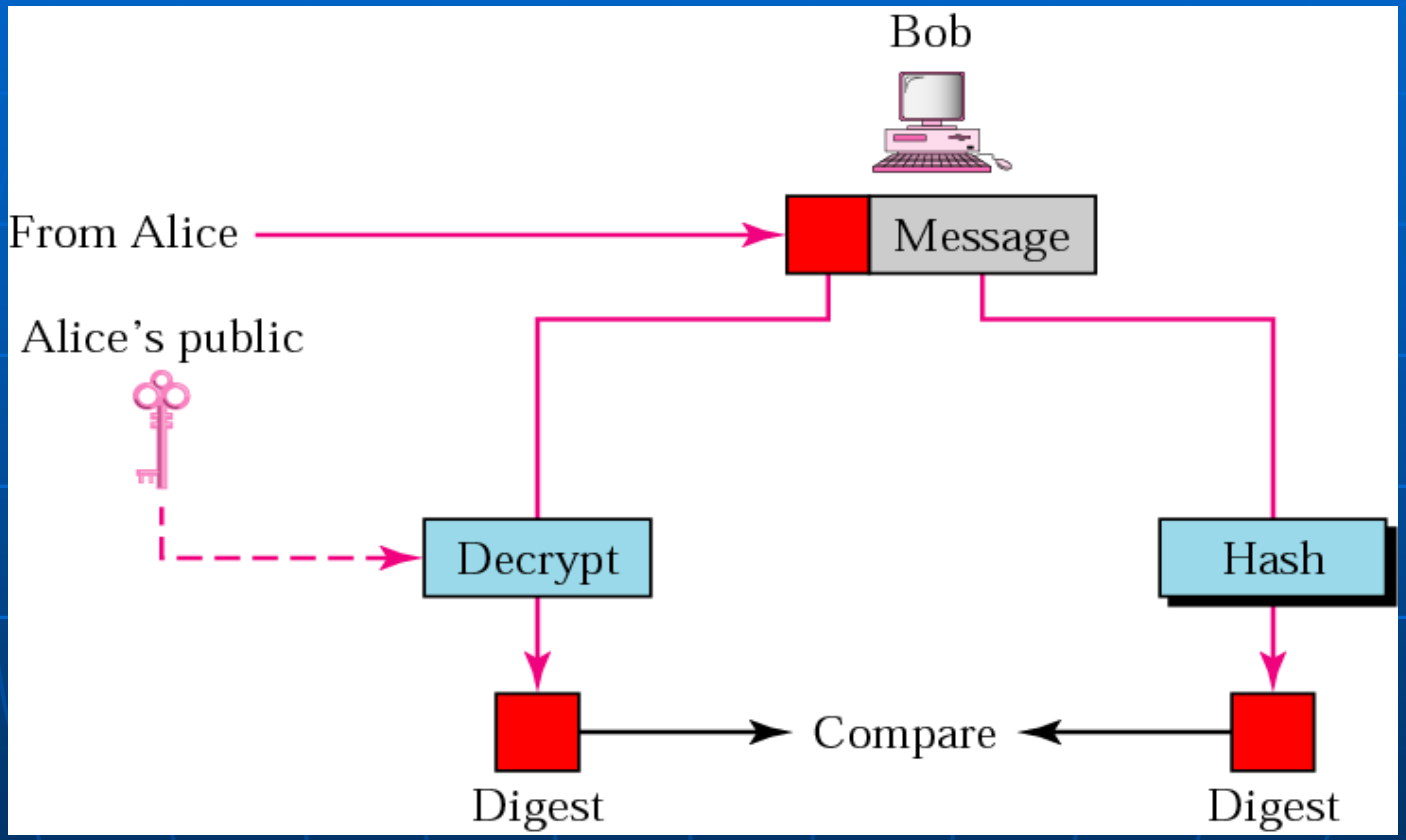


Figure 30.7 Receiver site



30.3 User Authentication

With Symmetric Key

With Public Key

Figure 30.8 Using a symmetric key only

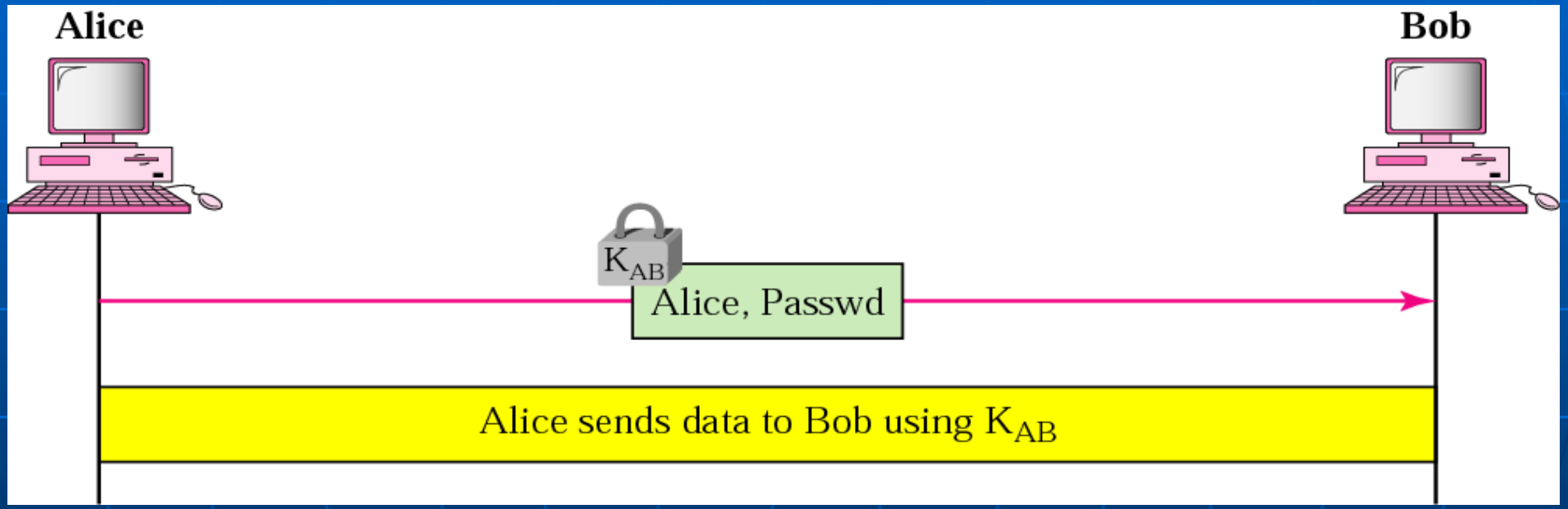


Figure 30.9 Using a nonce

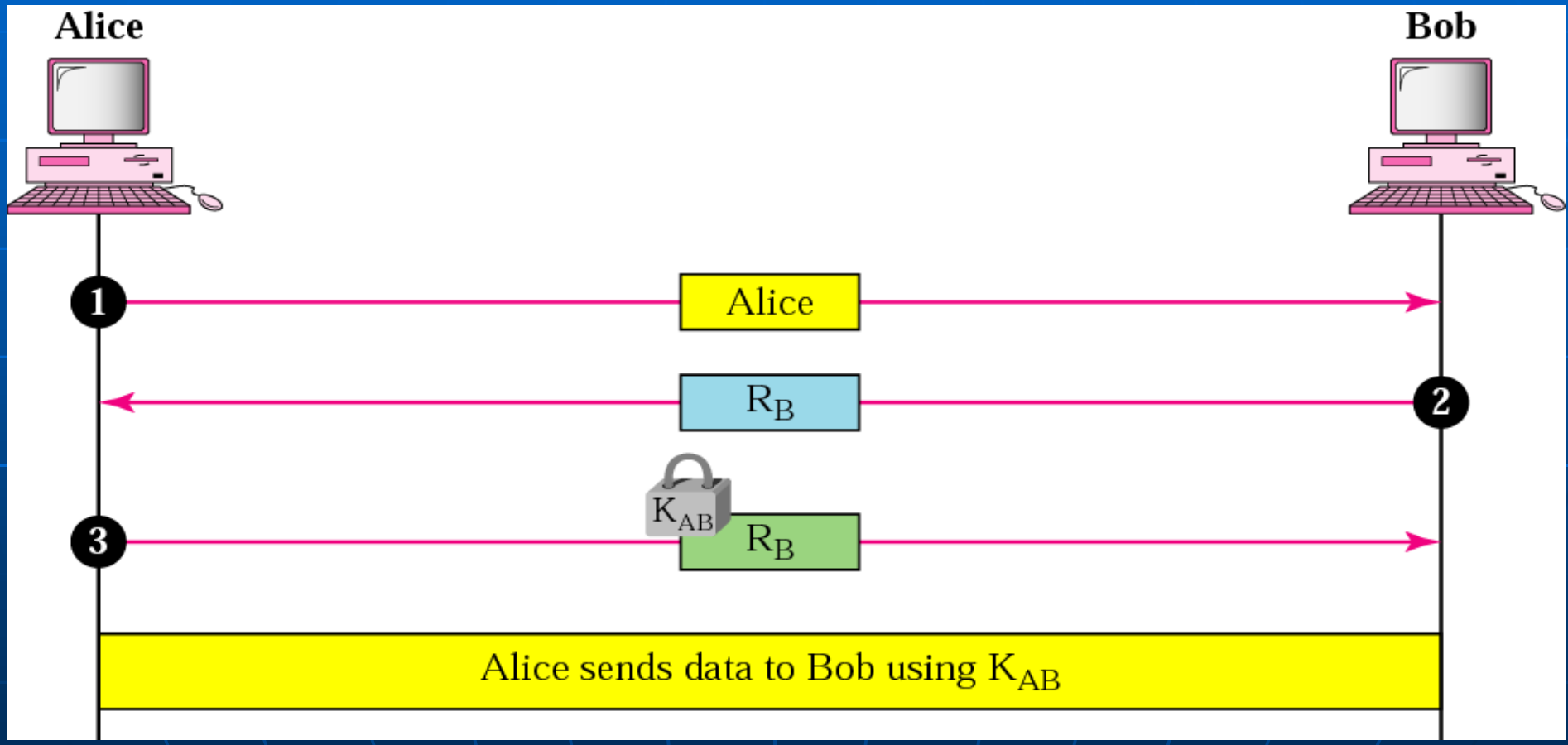
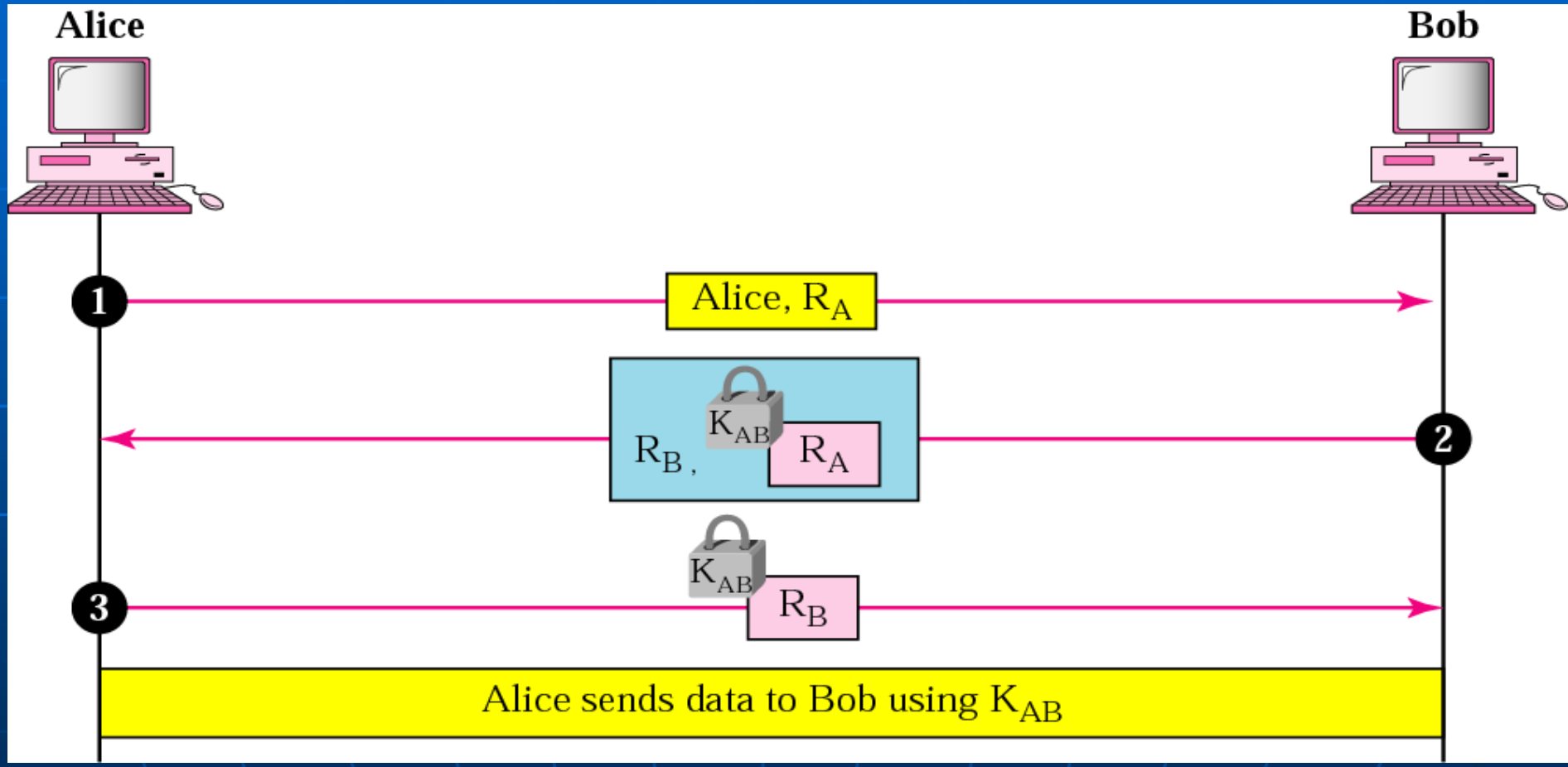


Figure 30.10 Bidirectional authentication



30.4 Key Management

Symmetric-Key Distribution

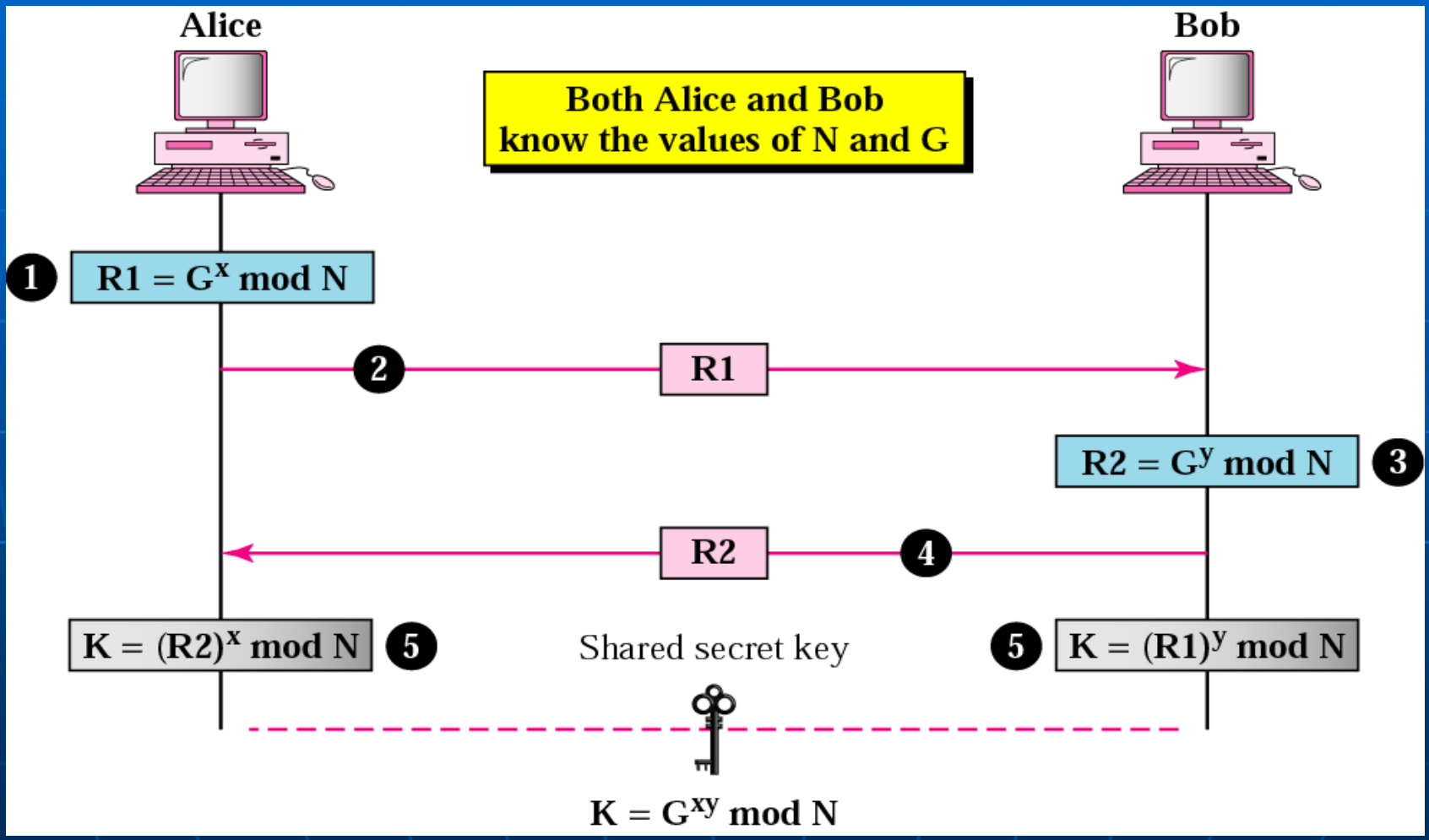
Public-Key Certification



Note:

A symmetric key between two parties is useful if it is used only once; it must be created for one session and destroyed when the session is over.

Figure 30.11 Diffie-Hellman method





Note:

The symmetric (shared) key in the Diffie-Hellman protocol is $K = G^{xy} \bmod N$.

Example 2

Assume G = 7 and N = 23. The steps are as follows:

1. Alice chooses x = 3 and calculates R1 = 7^3 mod 23 = 21.
2. Alice sends the number 21 to Bob.
3. Bob chooses y = 6 and calculates R2 = 7^6 mod 23 = 4.
4. Bob sends the number 4 to Alice.
5. Alice calculates the symmetric key K = 4^3 mod 23 = 18.
6. Bob calculates the symmetric key K = 21^6 mod 23 = 18.

The value of K is the same for both Alice and Bob;
G^{xy} mod N = 7^{18} mod 23 = 18.

Figure 30.12 Man-in-the-middle attack

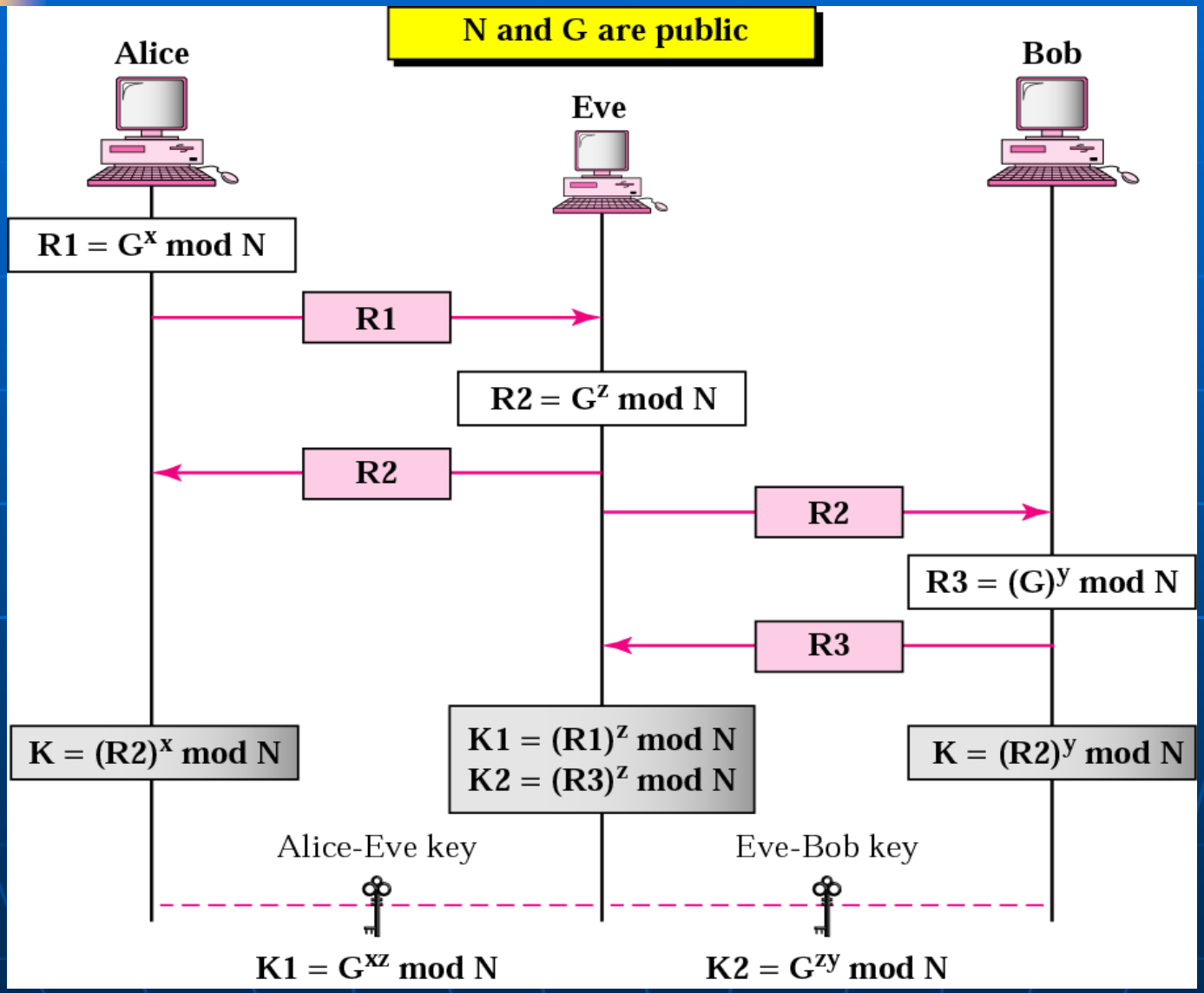


Figure 30.13 First approach using KDC

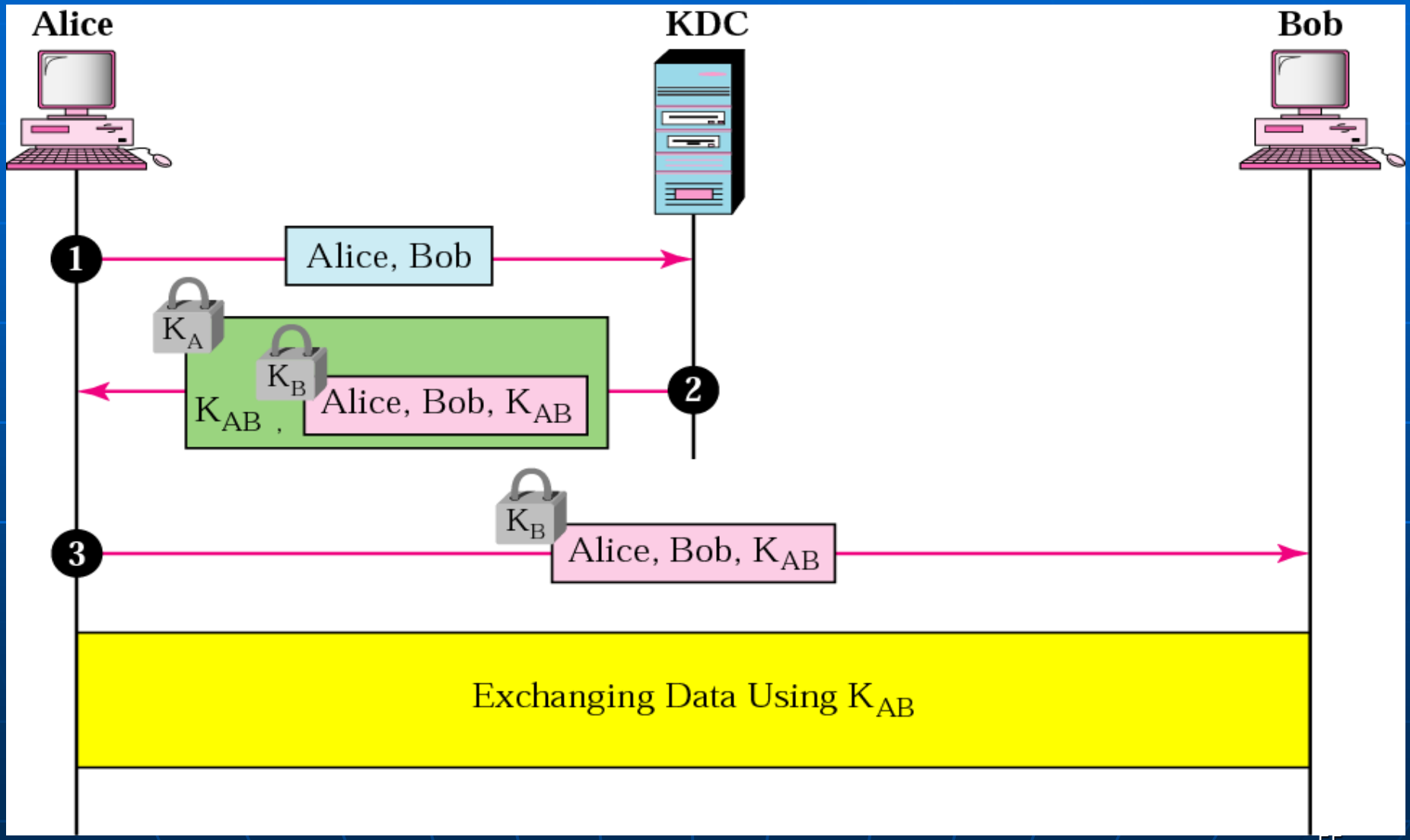


Figure 30.14 Needham-Schroeder protocol

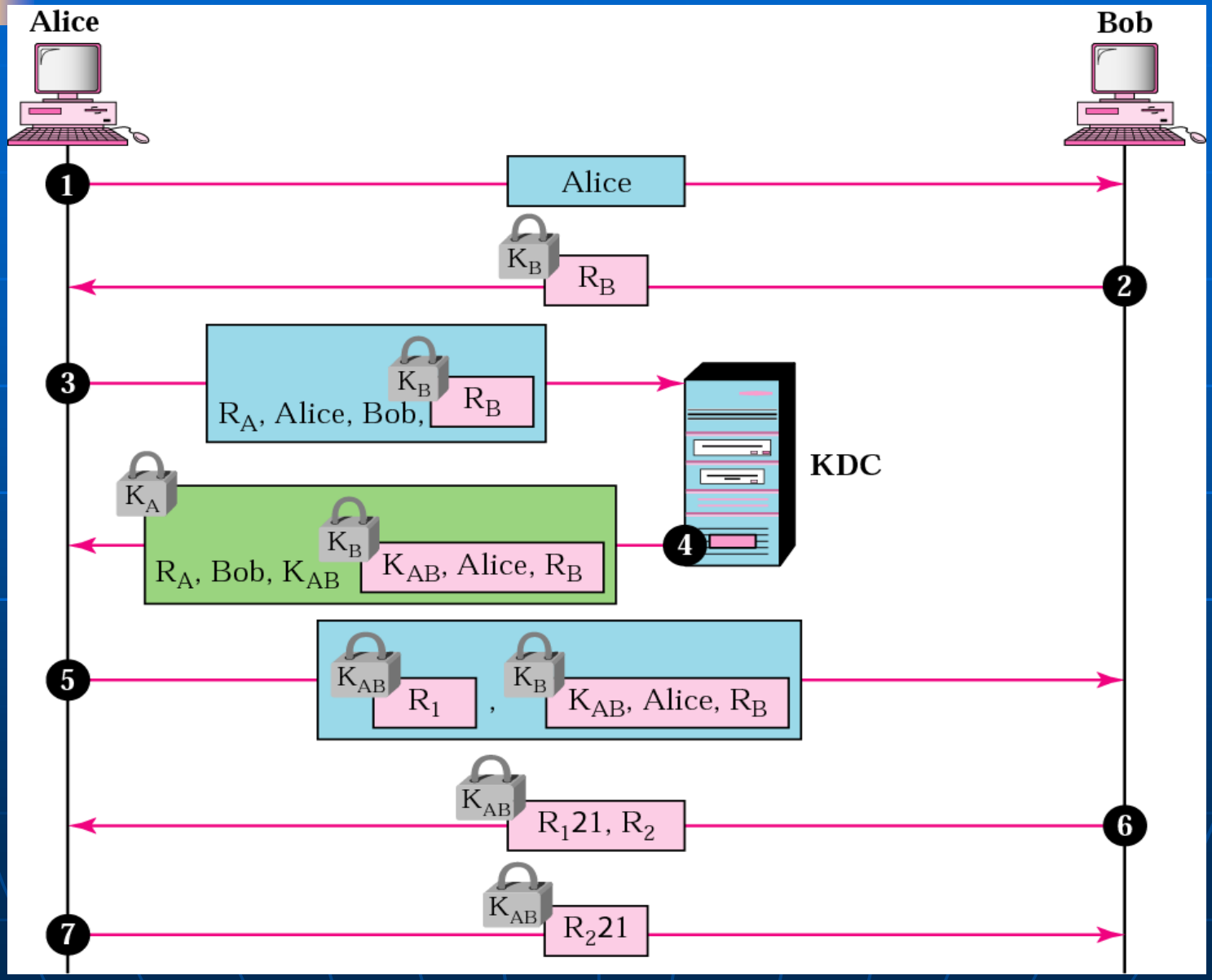
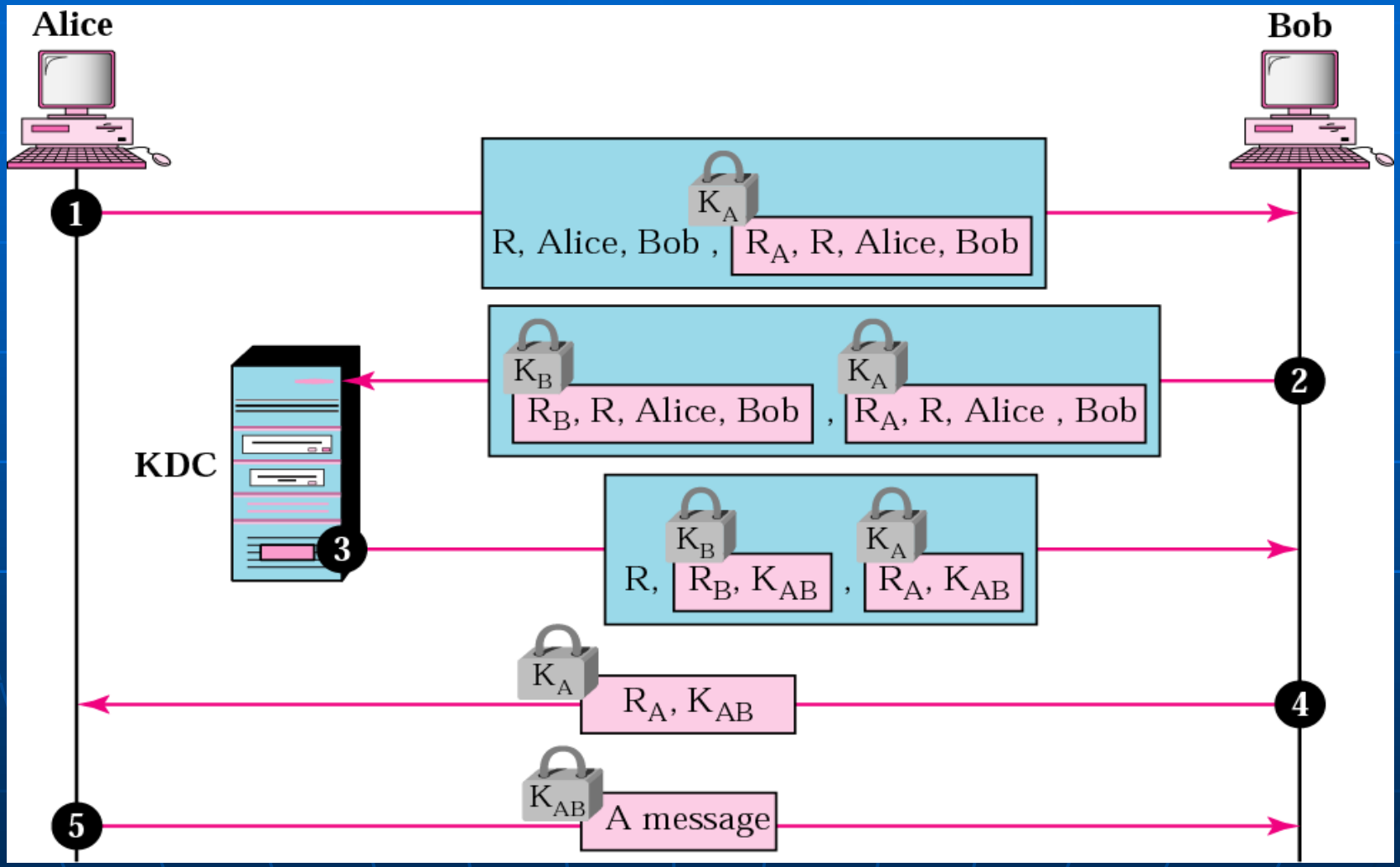


Figure 30.15 Otway-Rees protocol





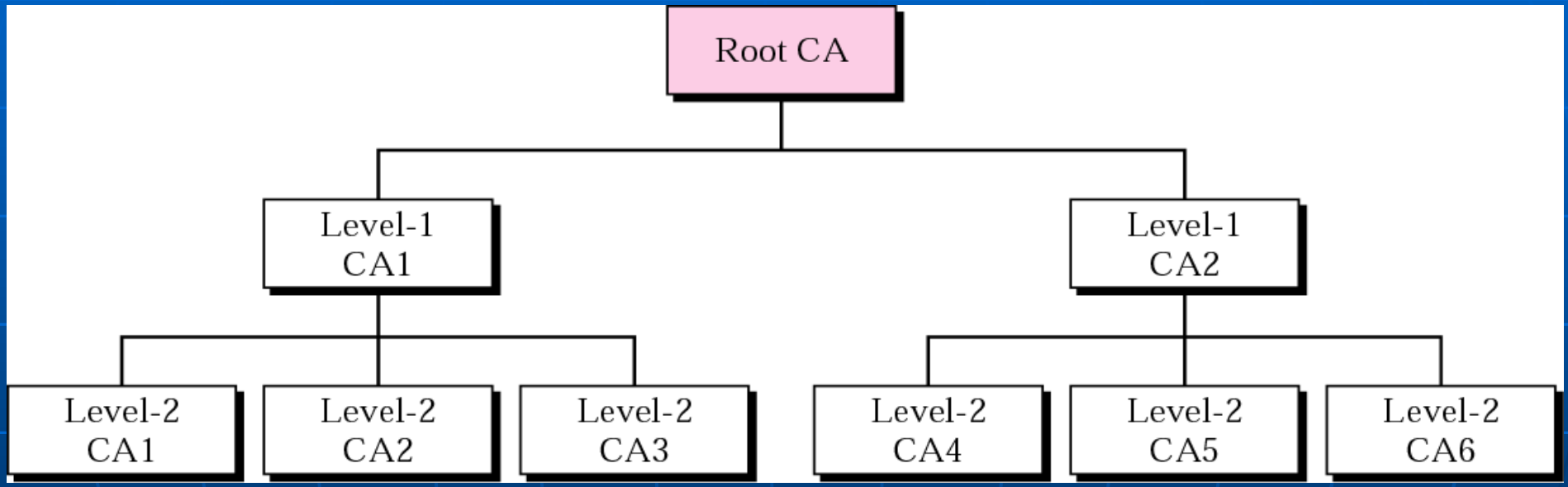
Note:

In public-key cryptography, everyone has access to everyone's public key.

Table 30.1 X.500 fields

Field	*Explanation*
Version	Version number of X.509
Serial number	The unique identifier used by the CA
Signature	The certificate signature
Issuer	The name of the CA defined by X.509
Validity period	Start and end period that certificate is valid
Subject name	The entity whose public key is being certified
Public key	The subject public key and the algorithms that use it

Figure 30.16 PKI hierarchy



30.5 Kerberos

Servers

Operation

Using Different Servers

Version 5

Realms

Figure 30.17 Kerberos servers

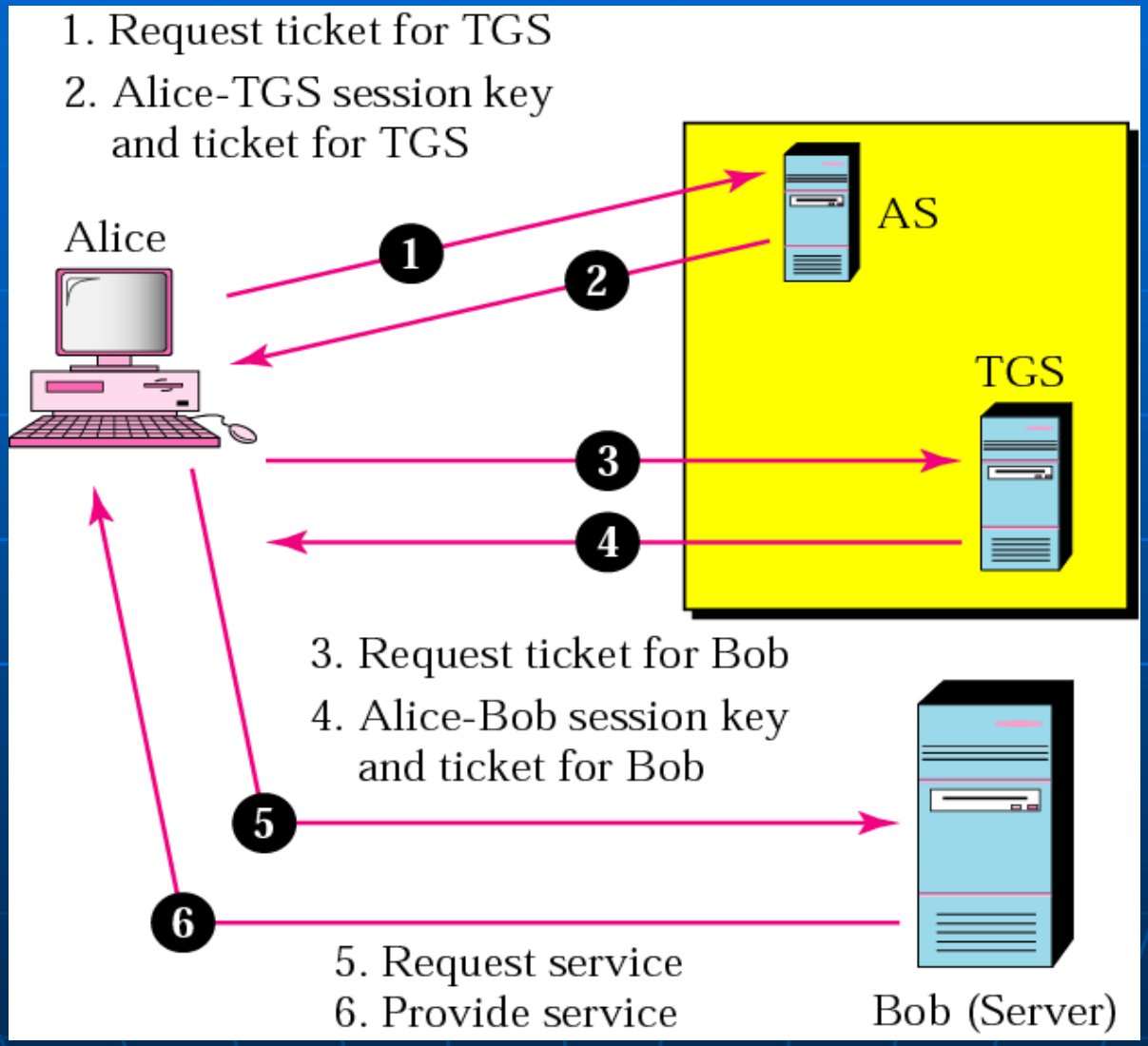
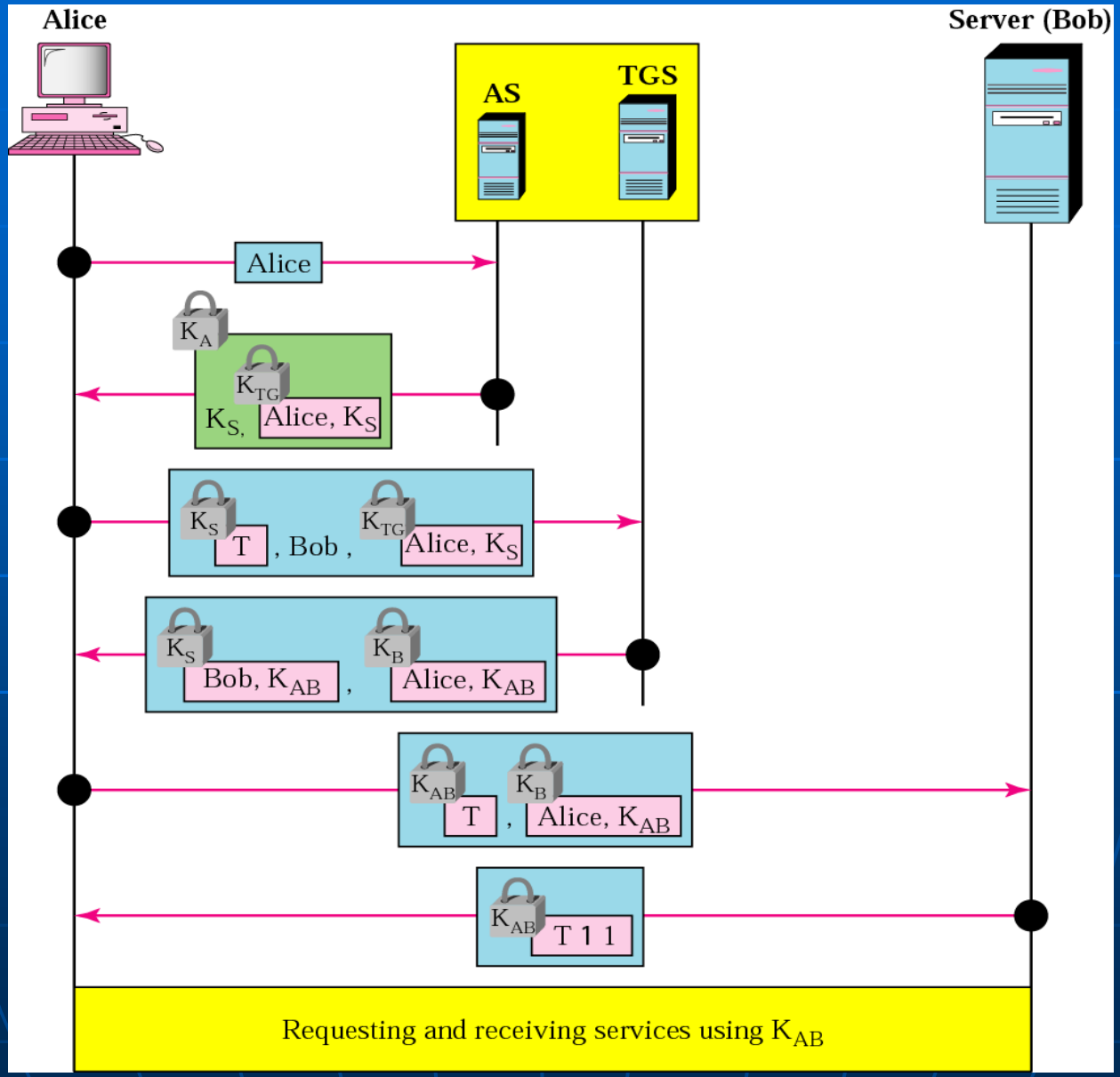


Figure 30.18 *Kerberos example*



Chapter 31

Security Protocols in the Internet

31.1 IP Level Security

Security Association

Two Modes

Two Security Protocols

Authentication Header (AH)

Encapsulating Security Payload (ESP)

Figure 31.1 Transport mode

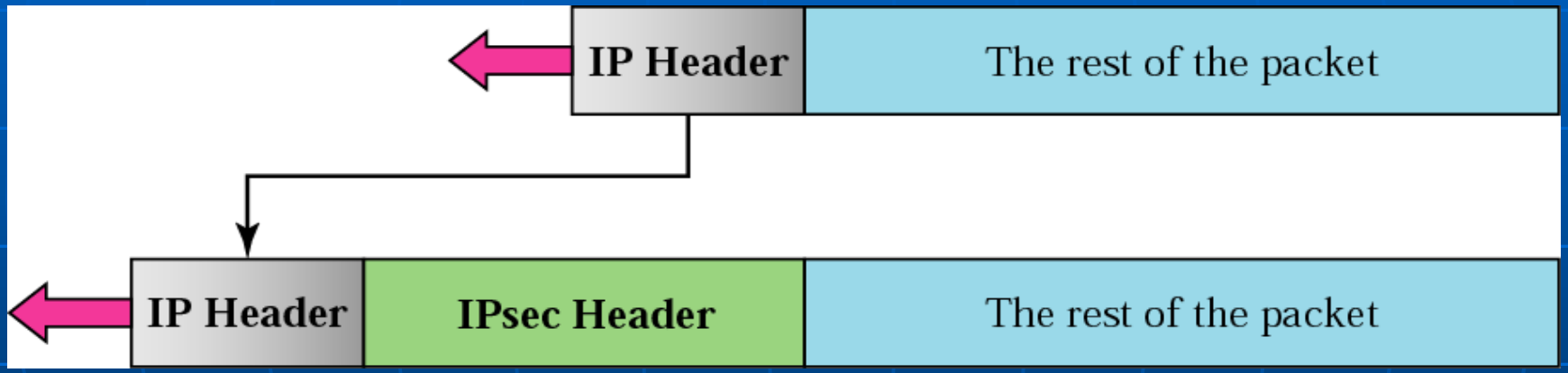


Figure 31.2 Tunnel mode

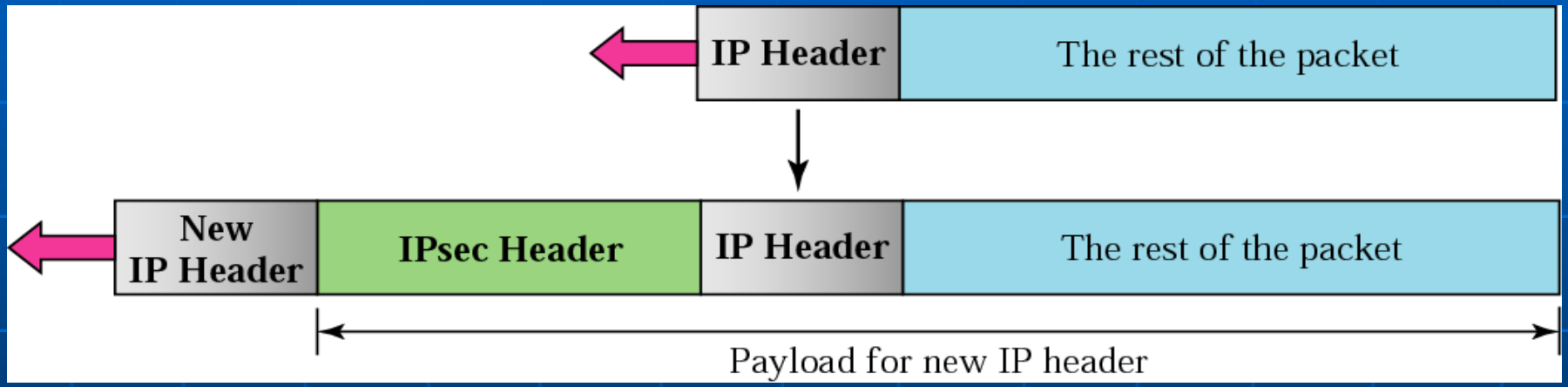
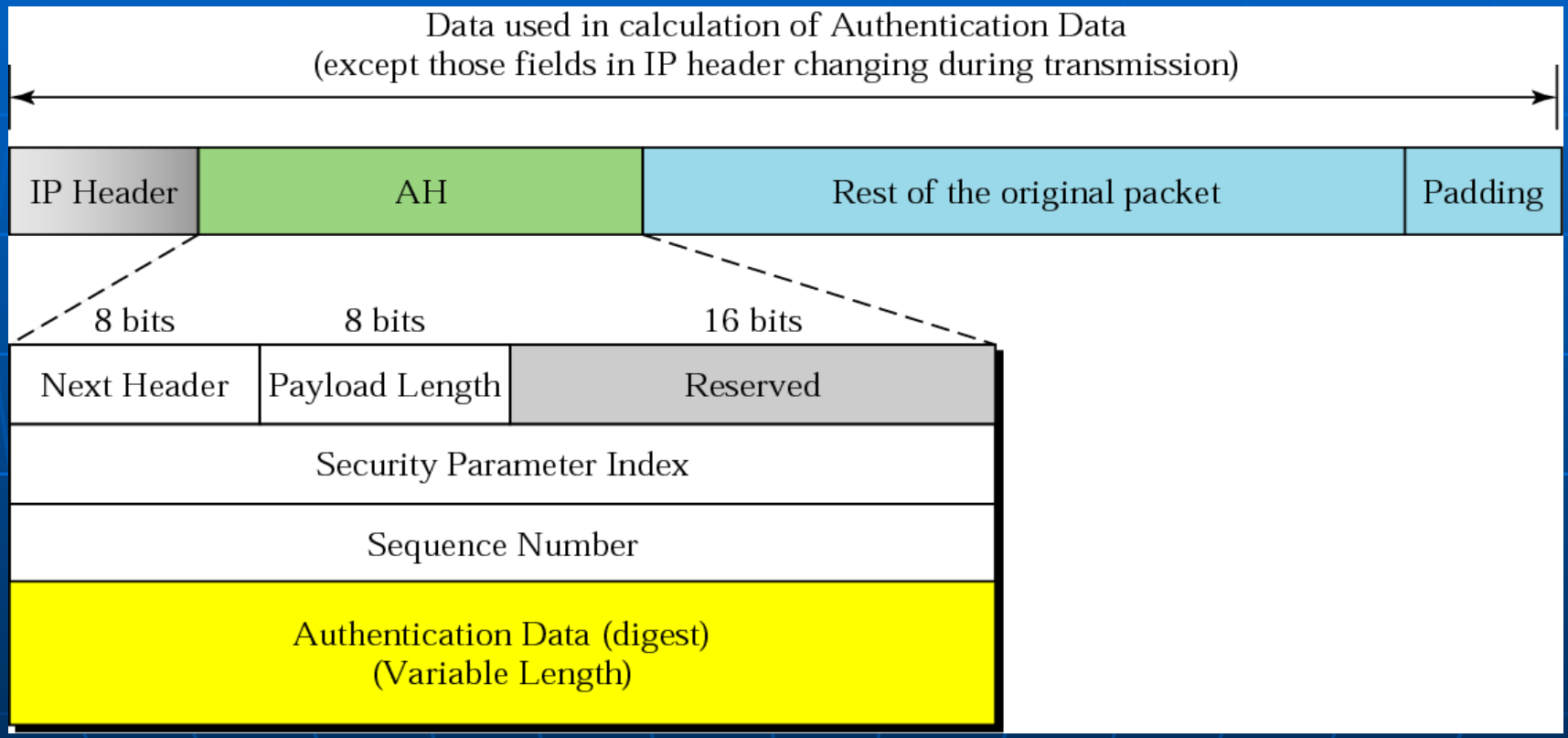


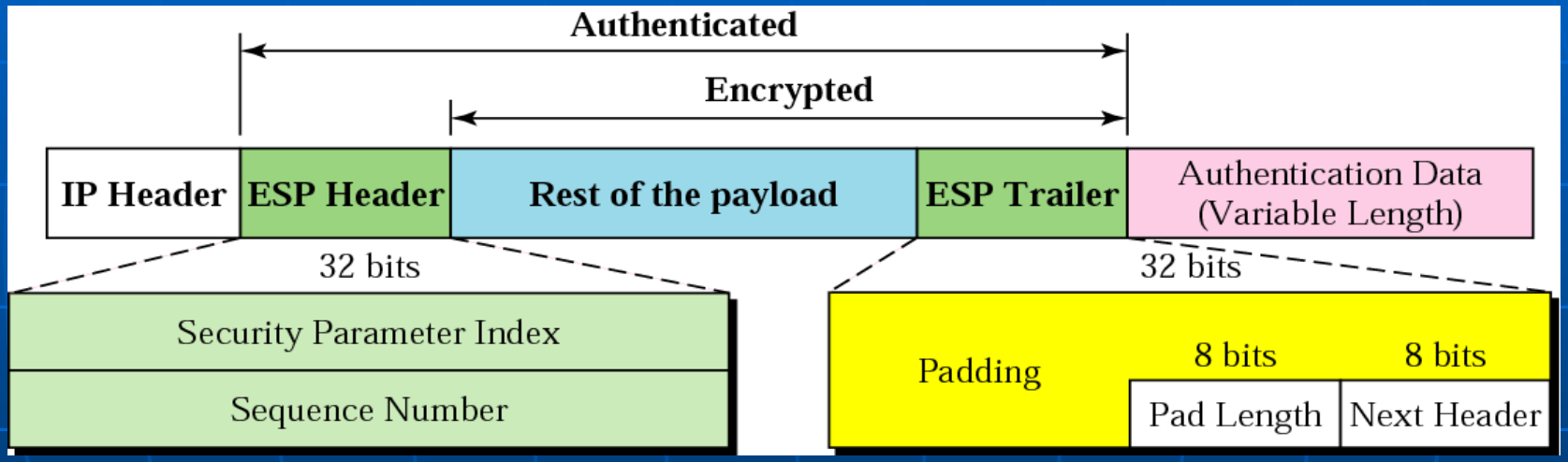
Figure 31.3 AH





The AH protocol provides source authentication and data integrity, but not privacy.

Figure 31.4 ESP





ESP provides source authentication, data integrity, and privacy.

31.2 Transport Layer Security

Position of TLS

Two Protocols

Figure 31.5 Position of TLS

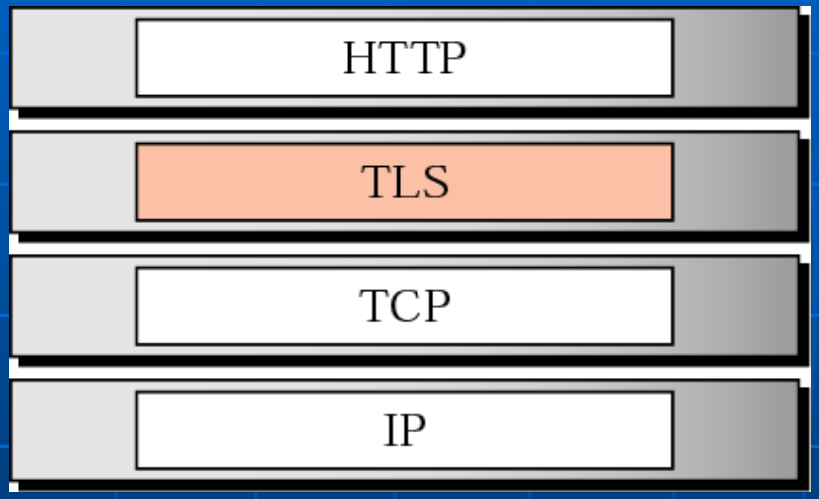
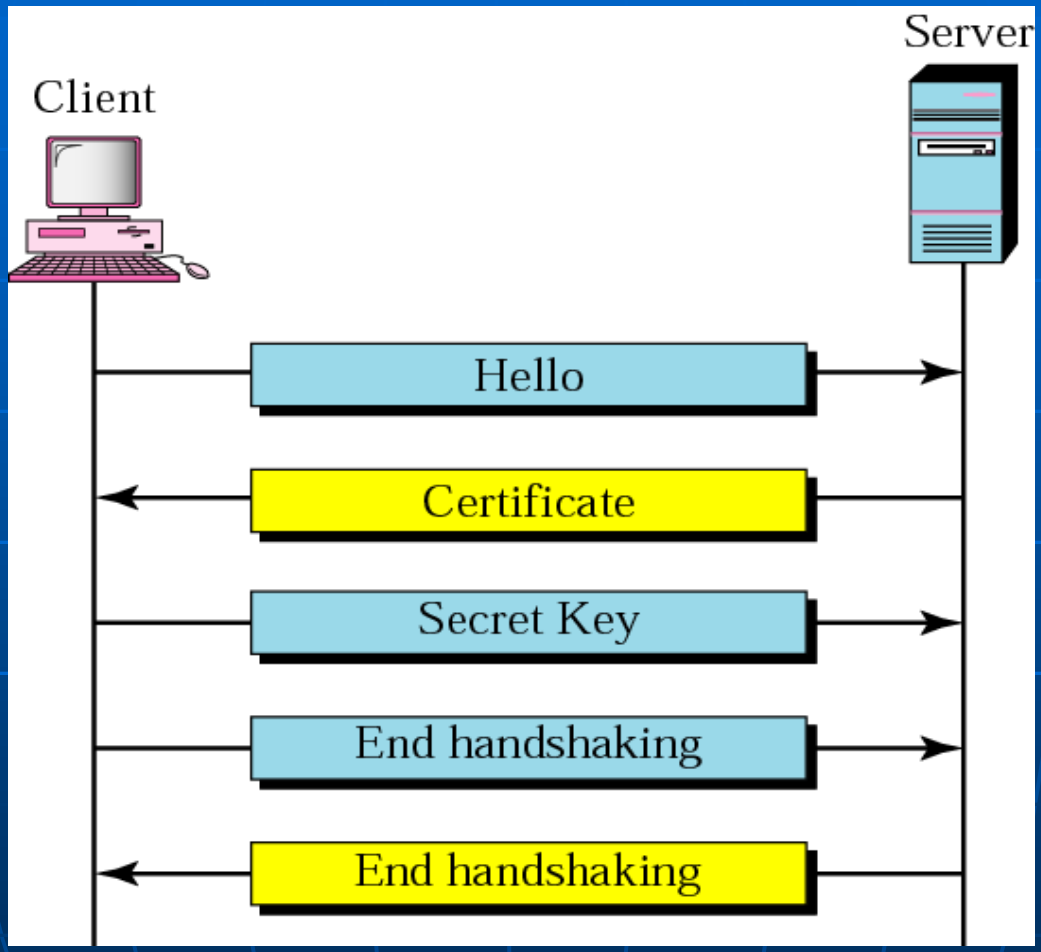


Figure 31.6 Handshake protocol



31.3 Application Layer Security

Pretty Good Privacy

Figure 31.7 PGP at the sender site

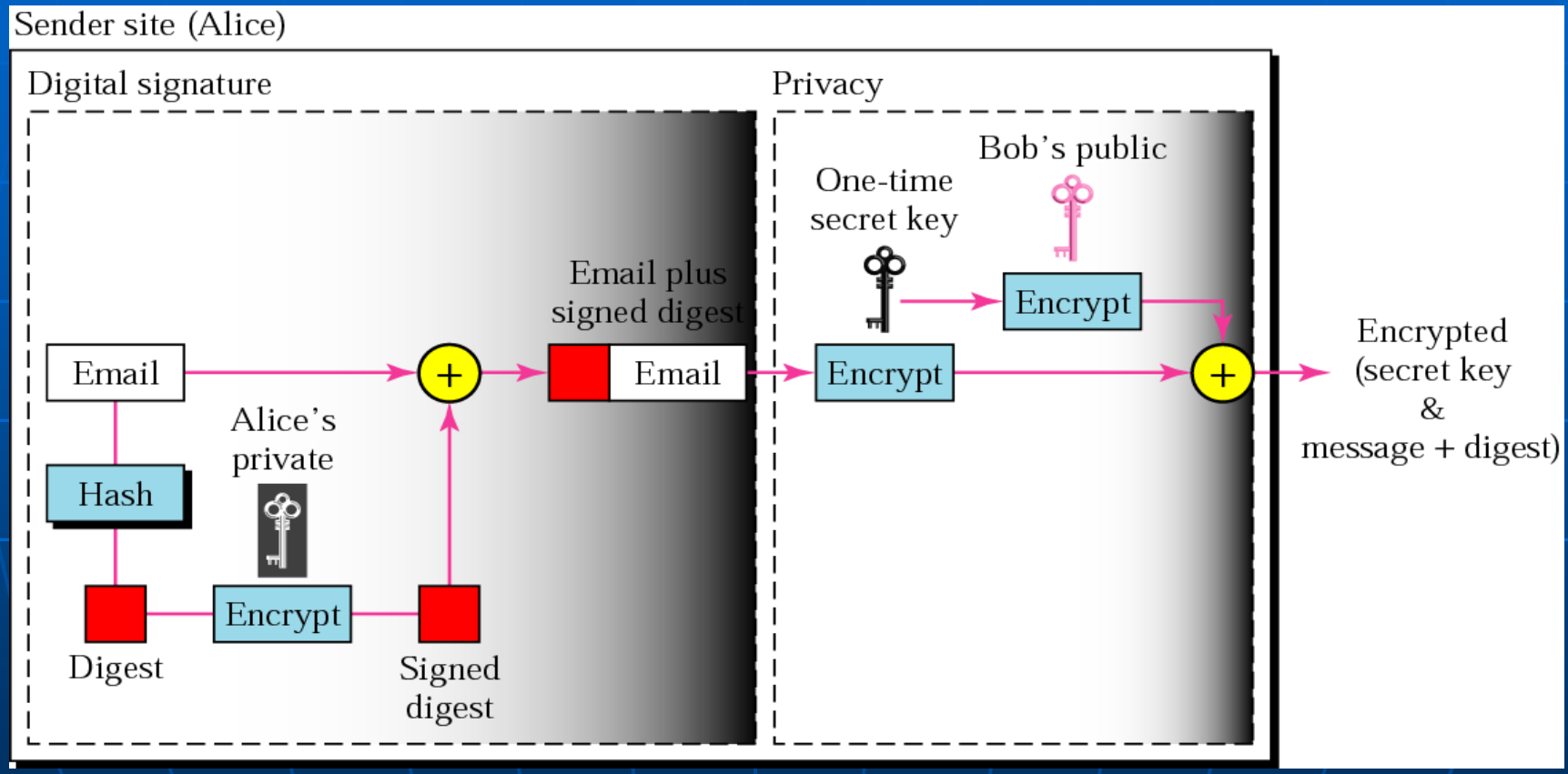
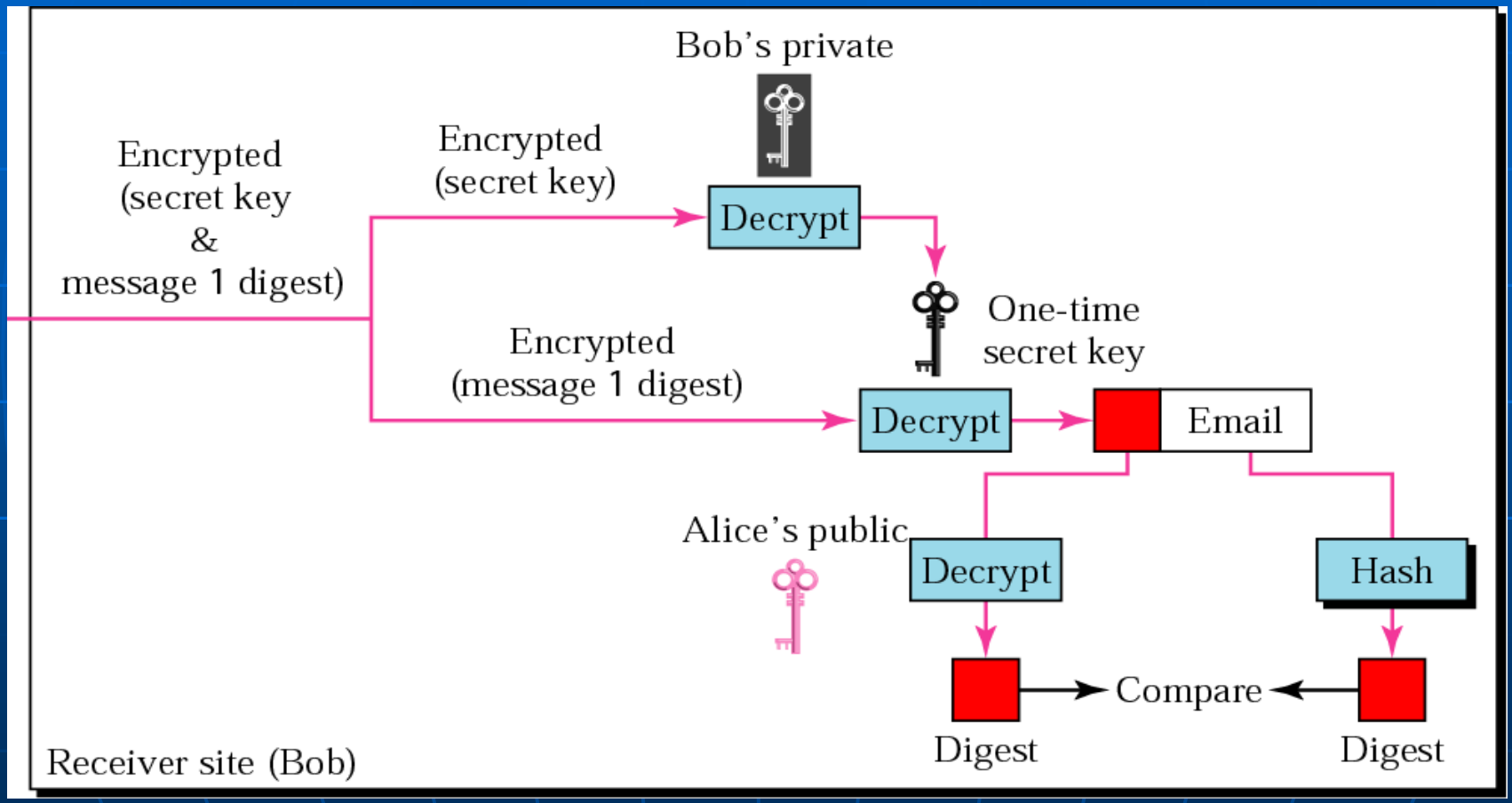


Figure 31.8 PGP at the receiver site



31.4 Firewalls

Packet-Filter Firewalls

Proxy Firewalls

Figure 31.9 Firewall

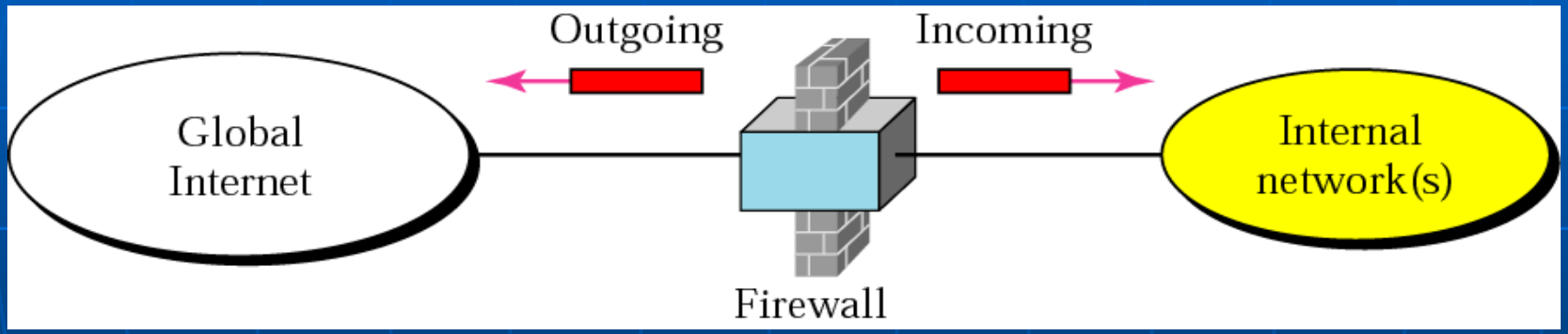
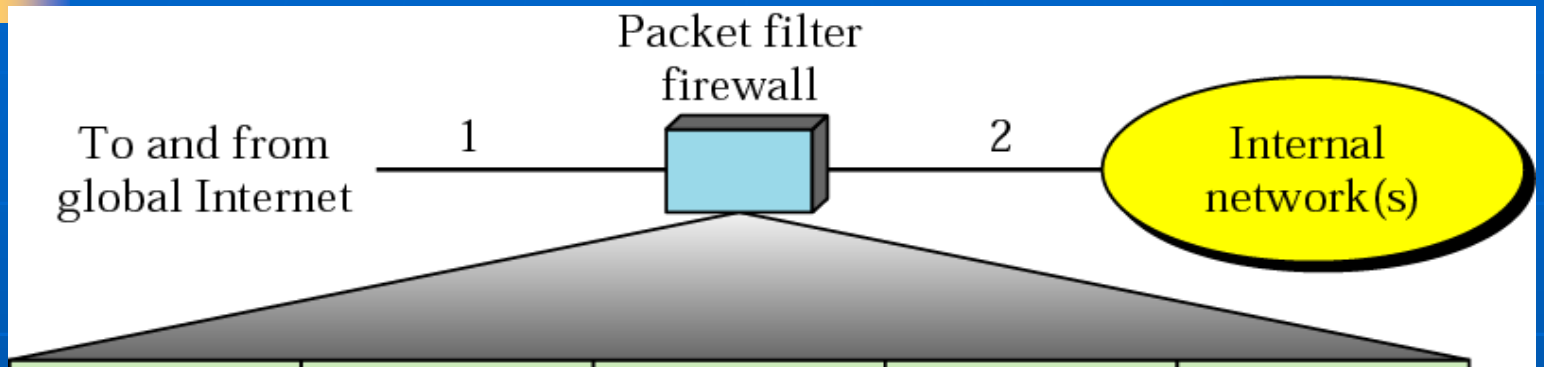


Figure 31.10 Packet-filter firewall



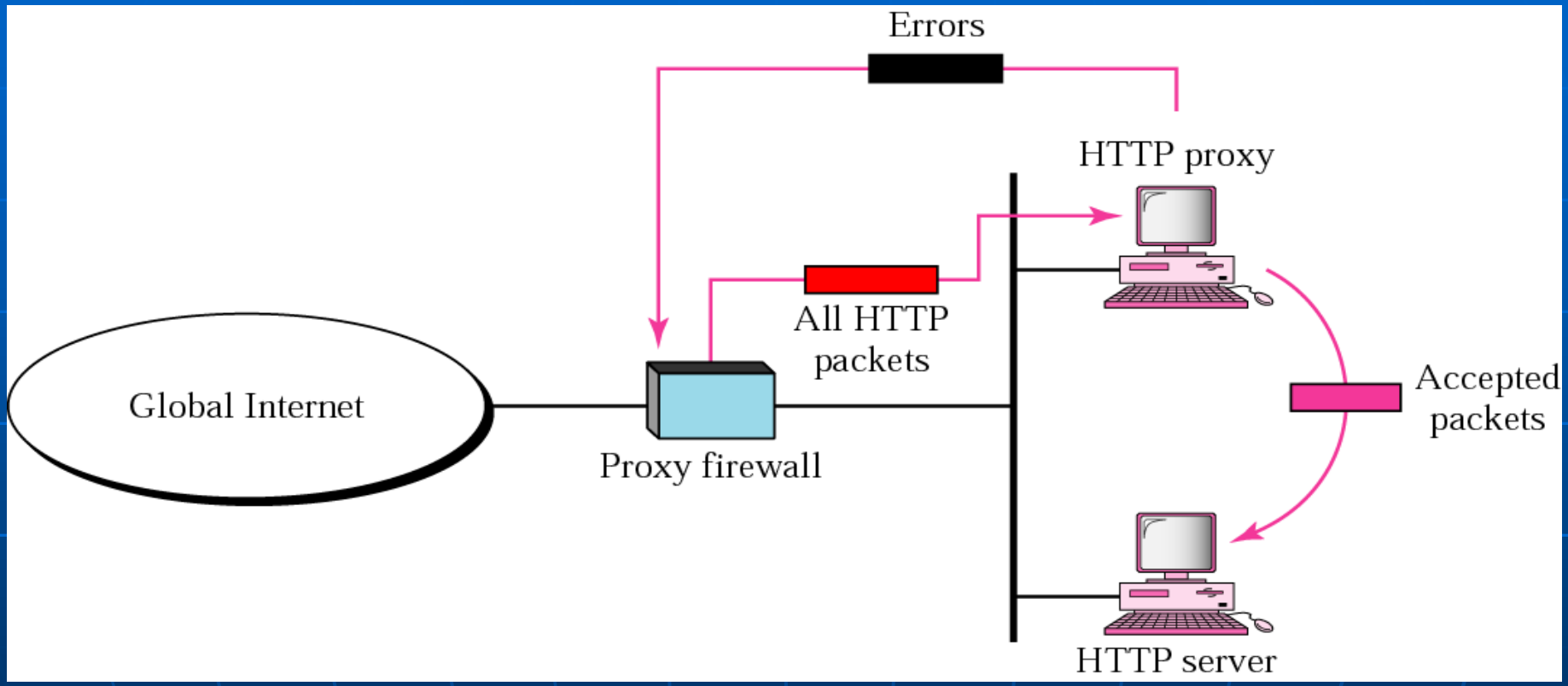
Interface	Source IP	Source Port	Destination IP	Destination Port
1	131.34.0.0	*	*	*
1	*	*	*	23
1	*	*	194.78.20.8	*
2	*	80	*	*



Note:

A packet-filter firewall filters at the network or transport layer.

Figure 31.11 Proxy firewall





Note:

A proxy firewall filters at the application layer.

31. 5 Virtual Private Networks

Private Networks

Achieving Privacy

VPN Technology

Table 31.1 Addresses for private networks

Prefix	Range			Total
10/8	10.0.0.0	to	10.255.255.255	2^{24}
172.16/12	172.16.0.0	to	172.31.255.255	2^{20}
192.168/16	192.168.0.0	to	192.168.255.255	2^{16}

Figure 31.12 Private network

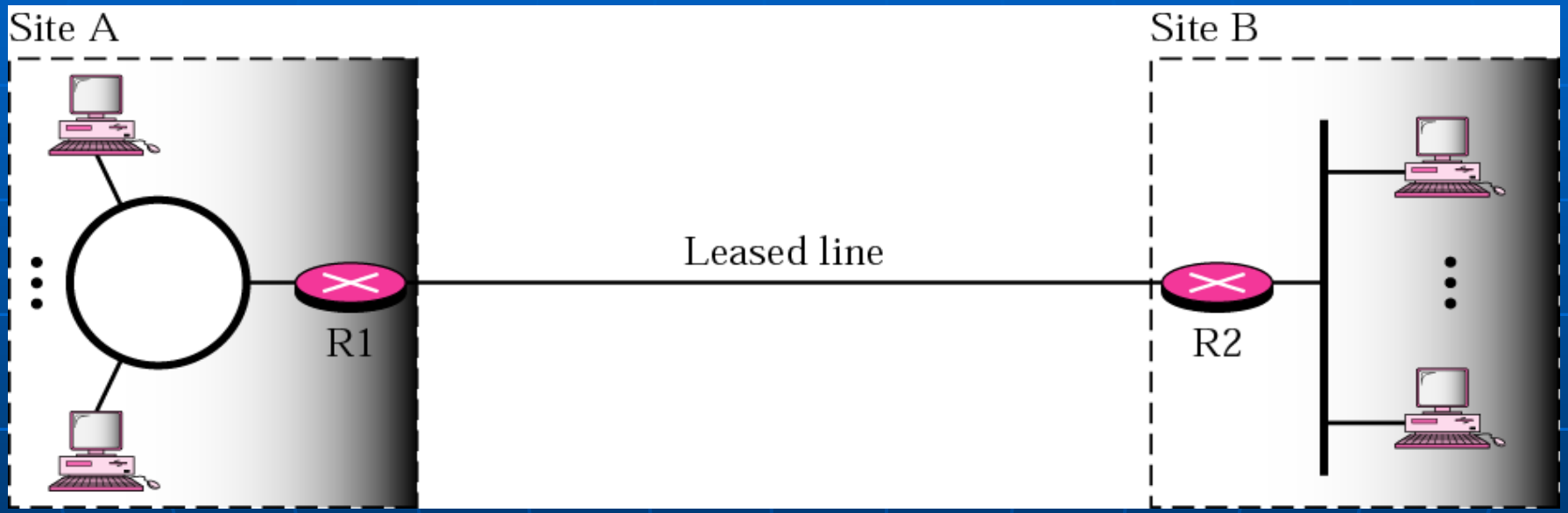


Figure 31.13 Hybrid network

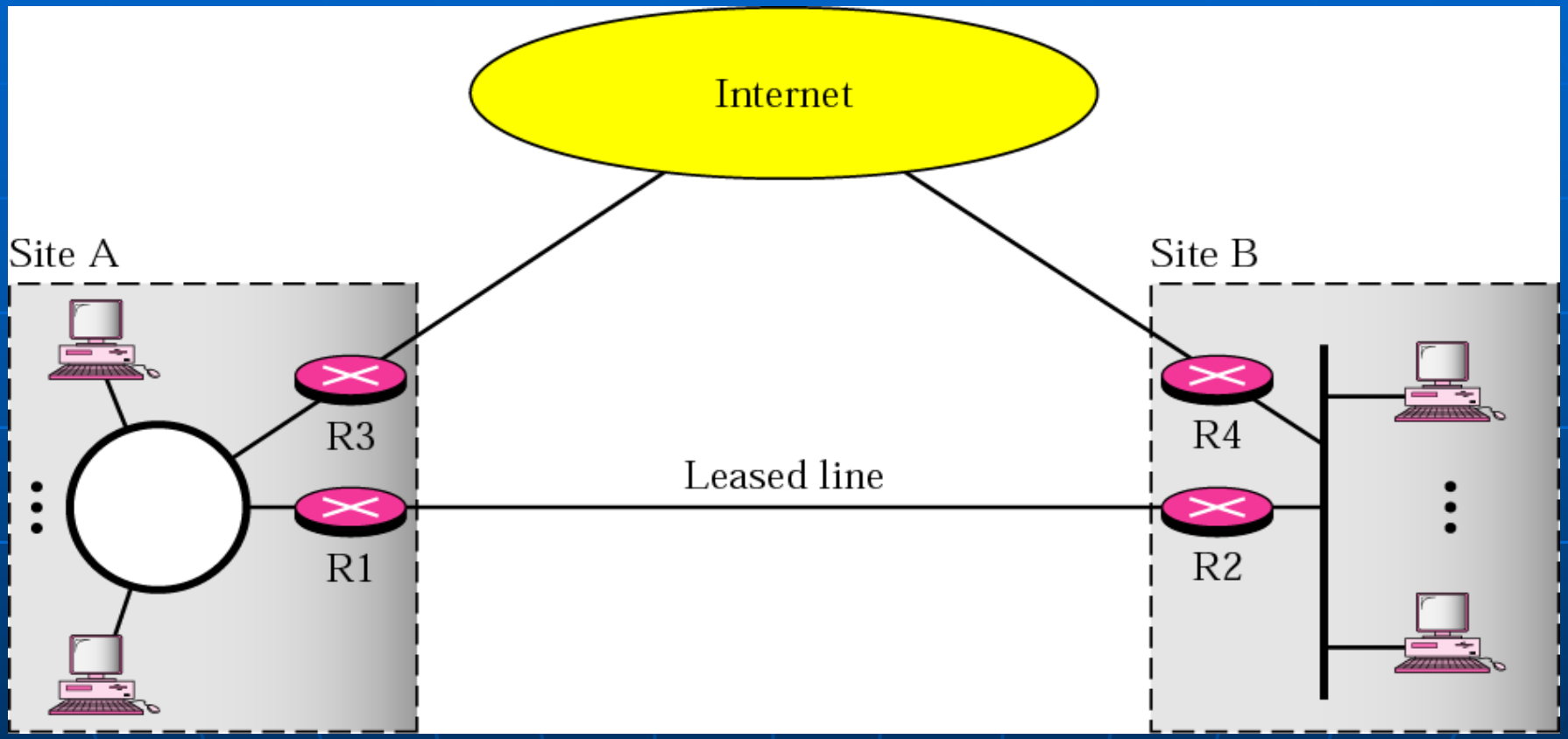


Figure 31.14 Virtual private network

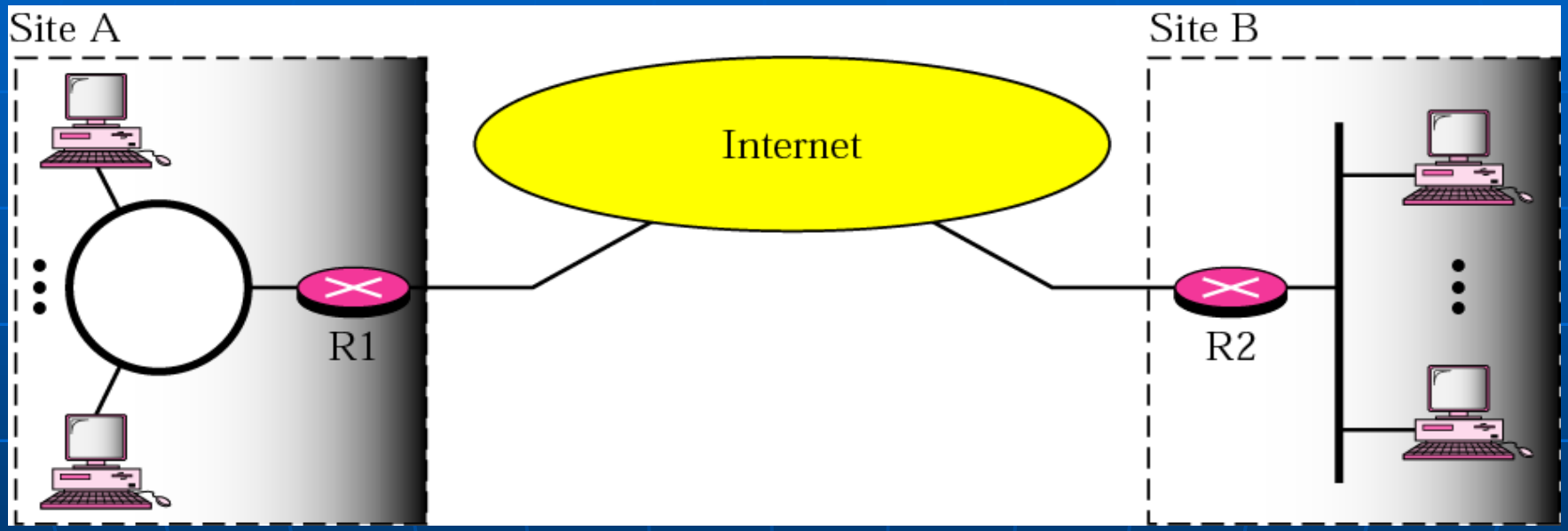


Figure 31.15 Addressing in a VPN

